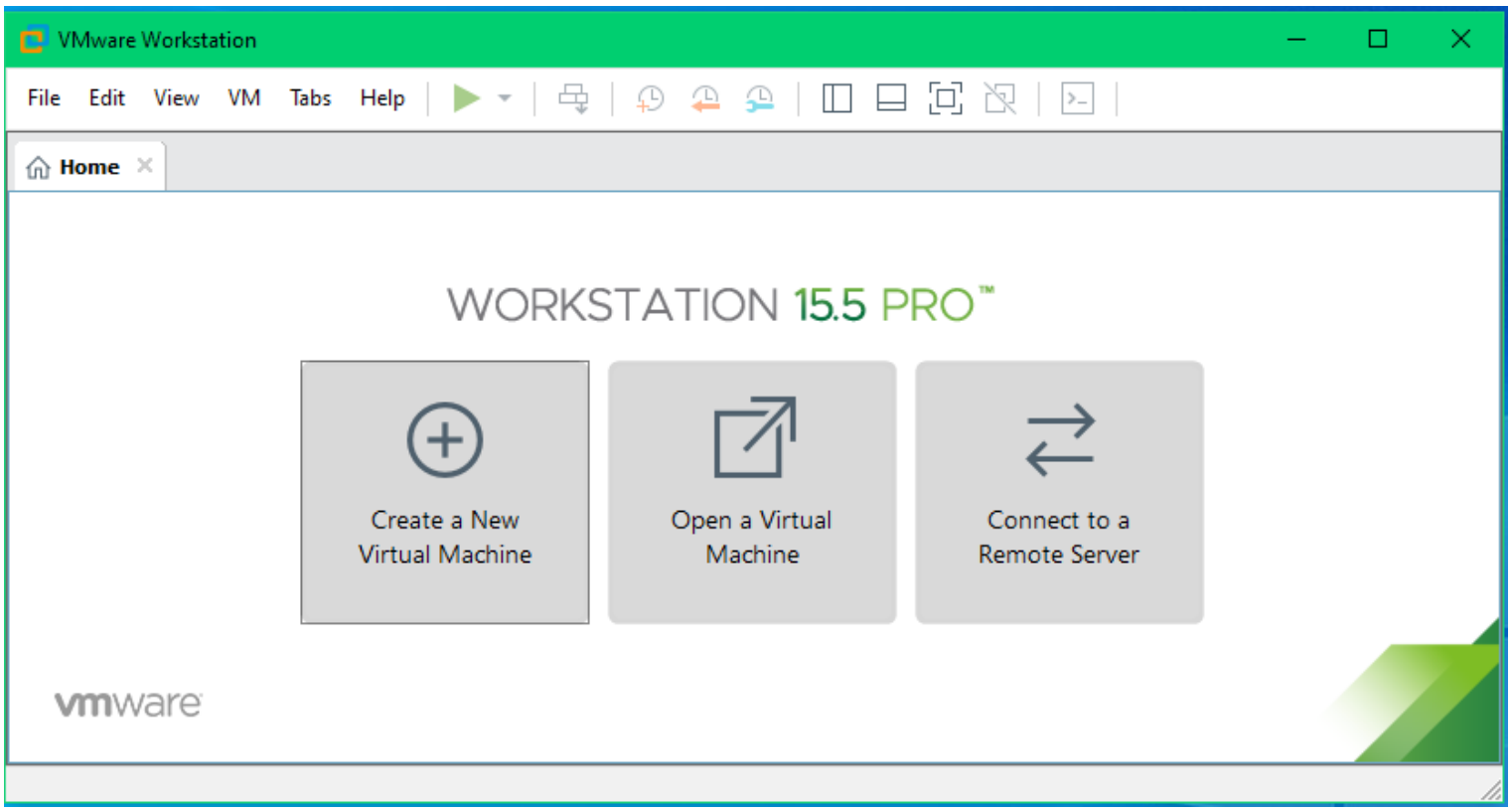
Virtual Machine Creation VMware Workstation - 21/01/2022

For this demonstration I will be creating a new virtual machine by firstly specifying my desired hardware specifications and networking options. Next I will install the Windows Server 2019 OS I previously downloaded on this virtual machine at the first boot. I will then Install VMware Tools before taking a snapshot of the newly created VM and that will conclude this tutorial.

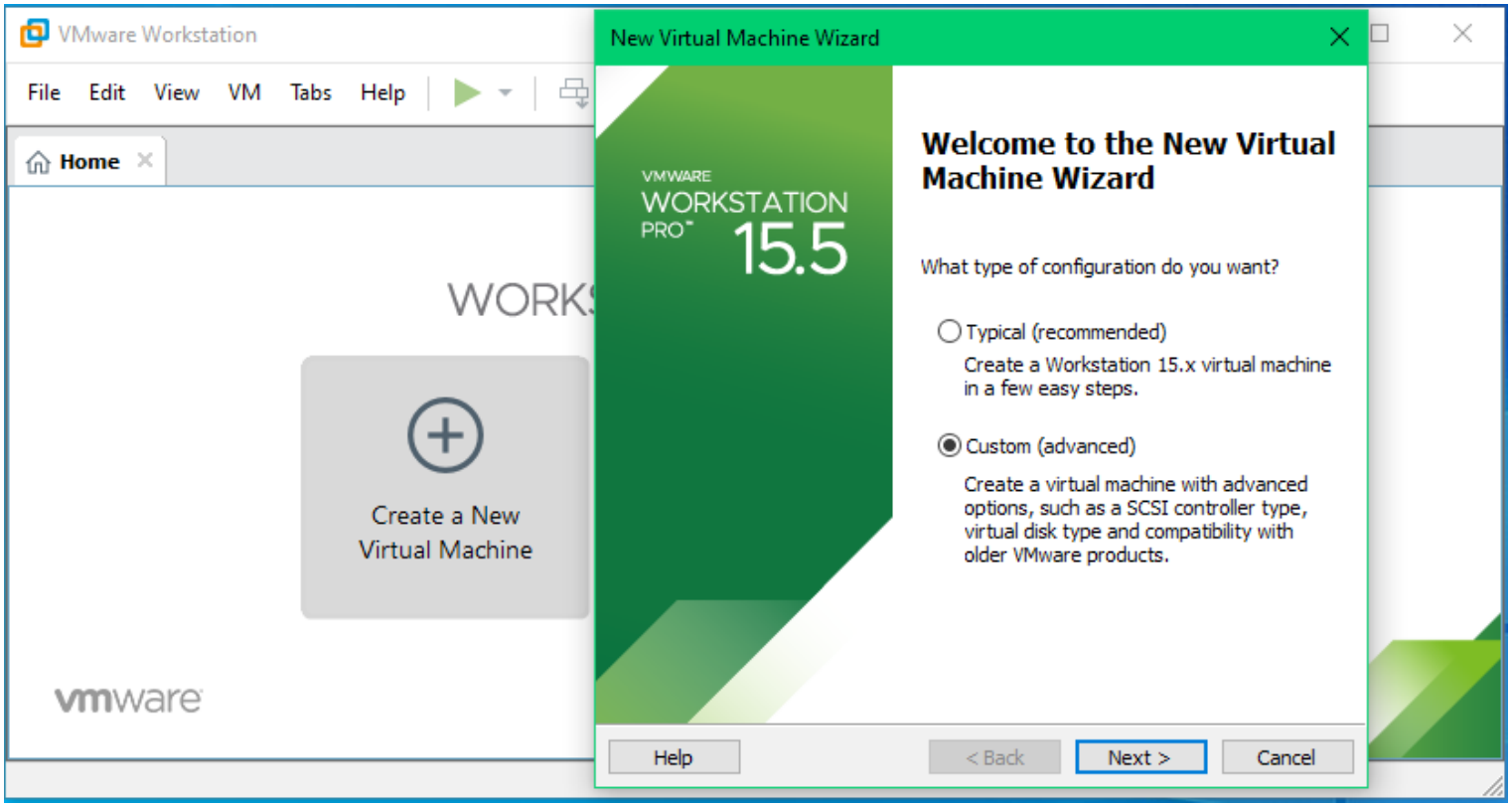
Note; regarding hypervisors (Virtualisation platforms), I use VMware Workstation as its a good product and I have a valid product key from my time at TAFESA however, there are many other free options out there.

1. Click Create A New Virtual Machine



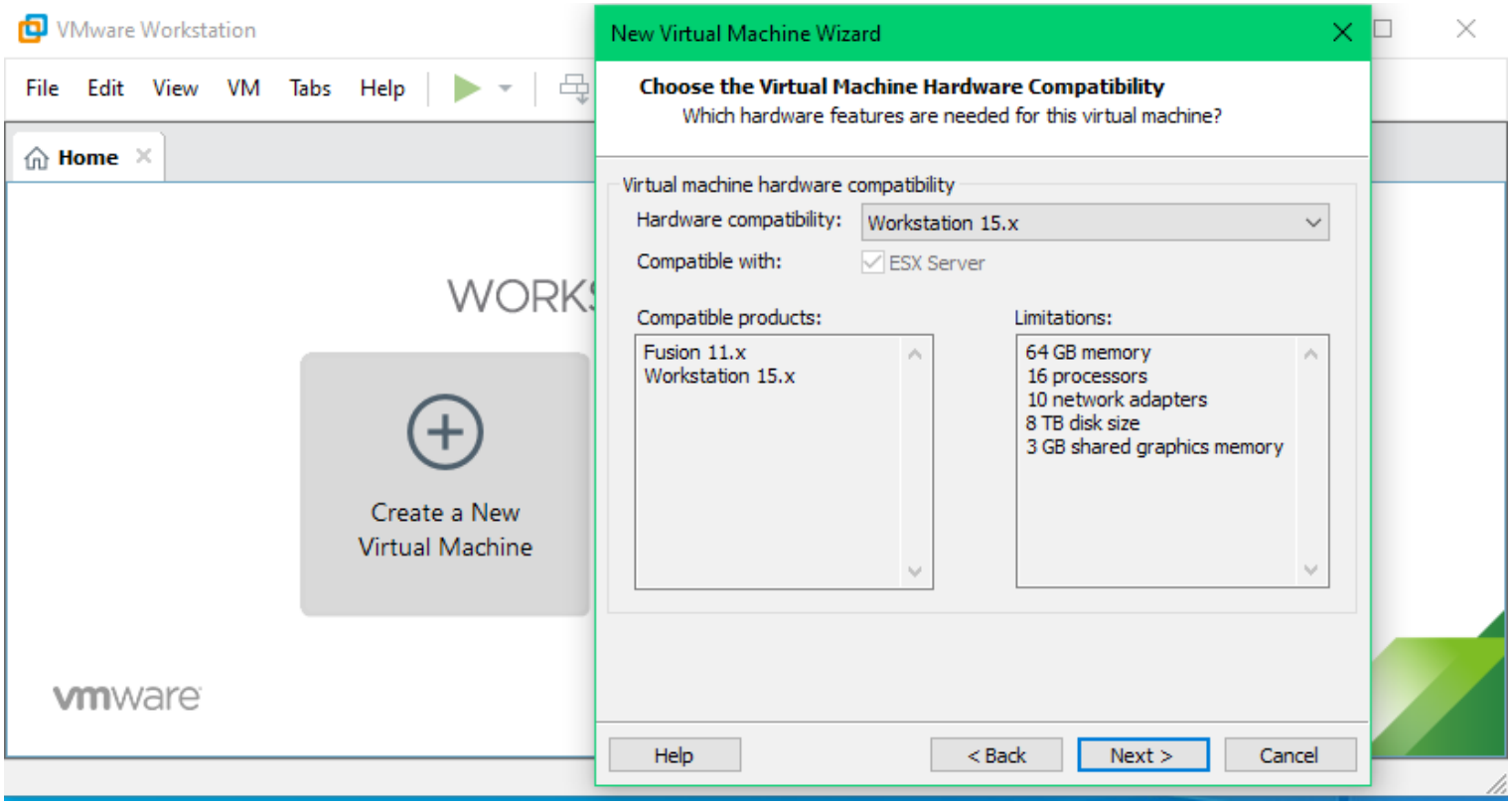
From the VMware Workstation Home-screen select the create a new virtual machine option. Or by heading to the file menu and selecting create a new virtual machine.

2. Select Configuration Type



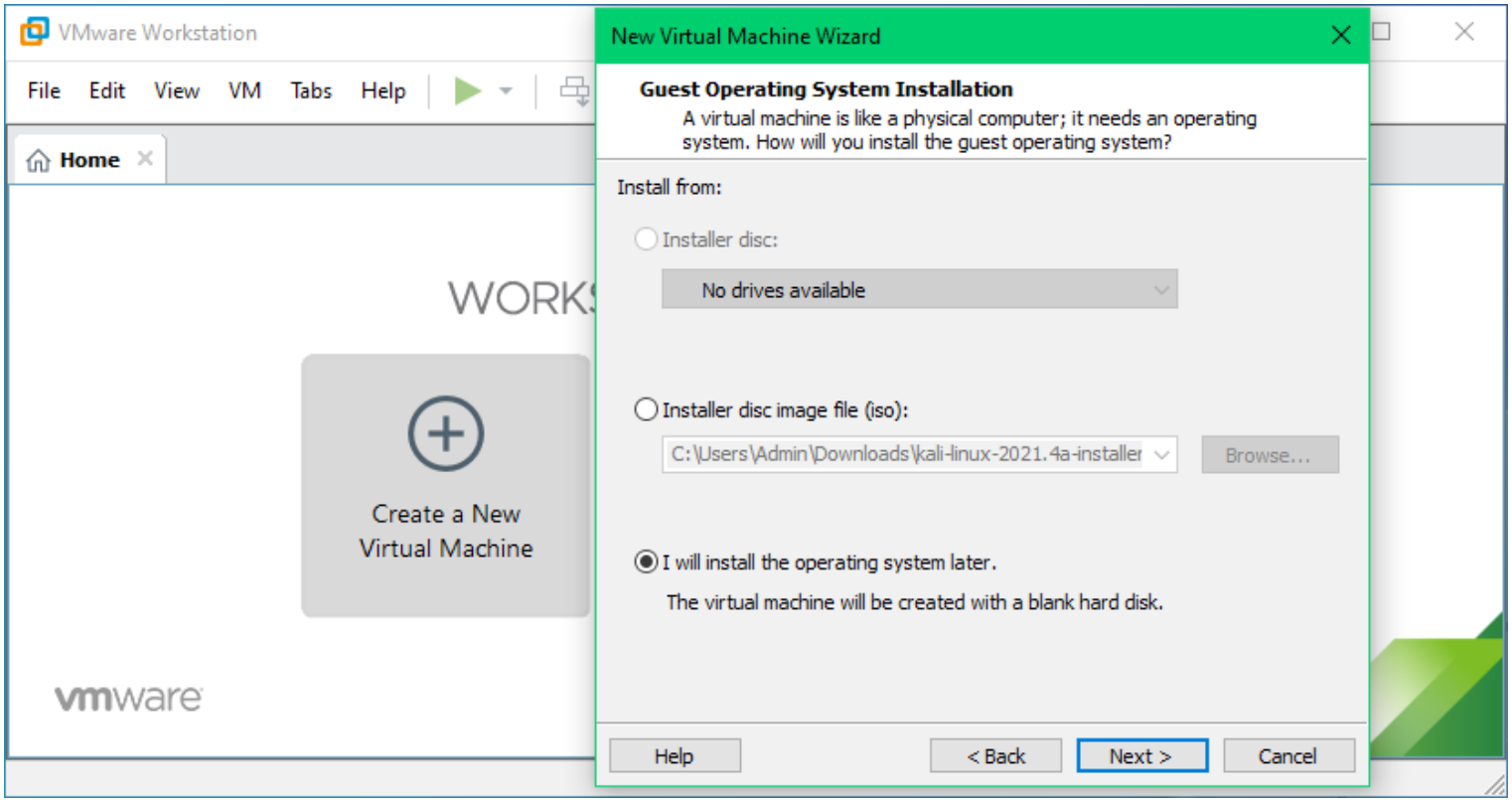
By selecting the custom radio button we will begin the advanced virtual machine wizard which allows us to customise several hardware and OS options. The typical method is fine, this is simply my personal preference when it comes to creating virtual machines.

3. Choose the hardware compatibility options



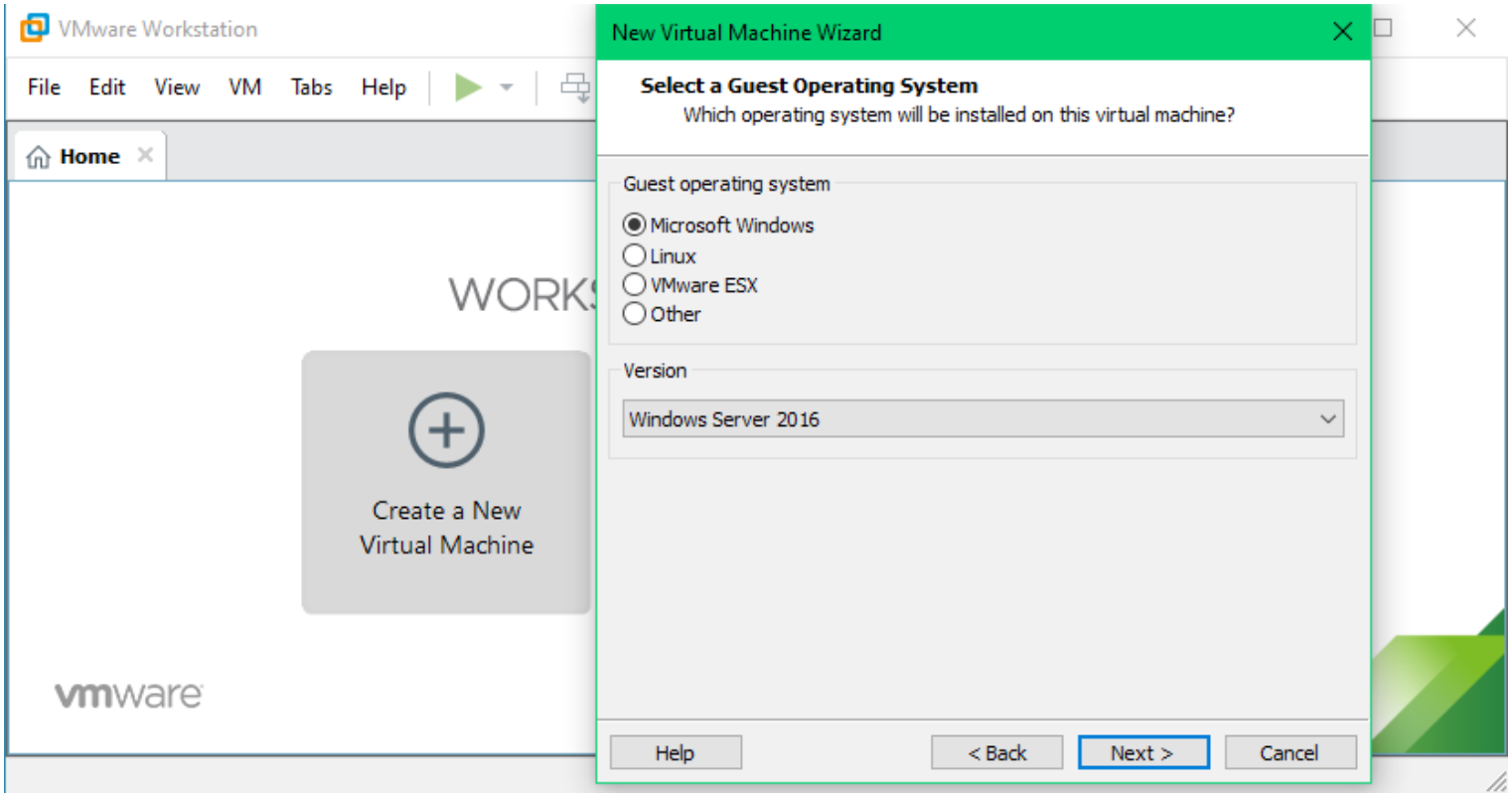
The hardware compatibility options can be used to allow for cross compatibility between other versions of VMware Workstation and other products. I simply accept the defaults as I only plan use Workstation 15.x.

4. Guest OS Installation



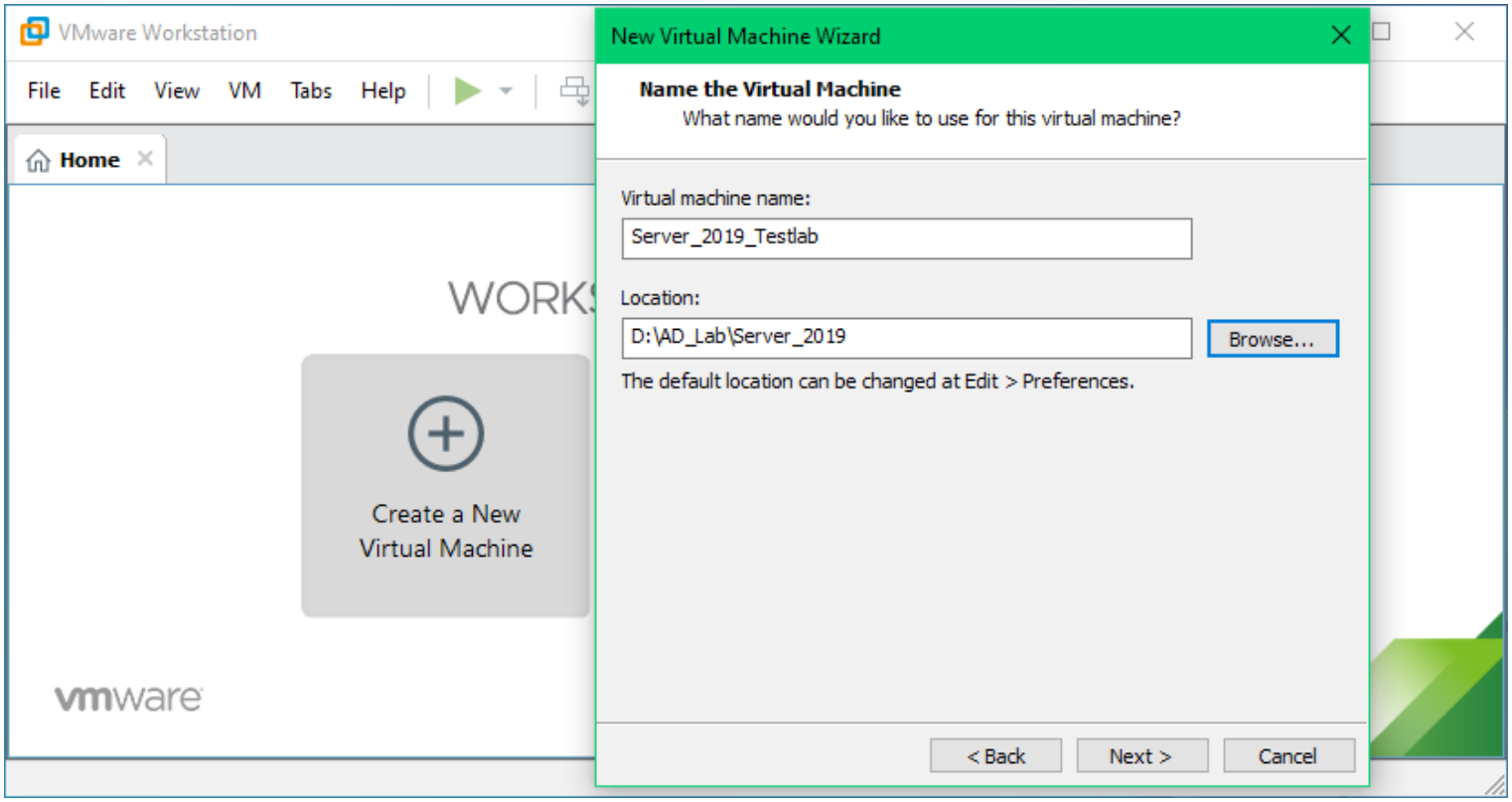
I have selected the option to install the operating system later. This is my preferred method to install a guest OS in VMware Workstation, You may prefer to simply select the Installer Disk Image File option (middle), browse and locate the ISO of the guest OS you wish to install.

5. Guest OS Selection



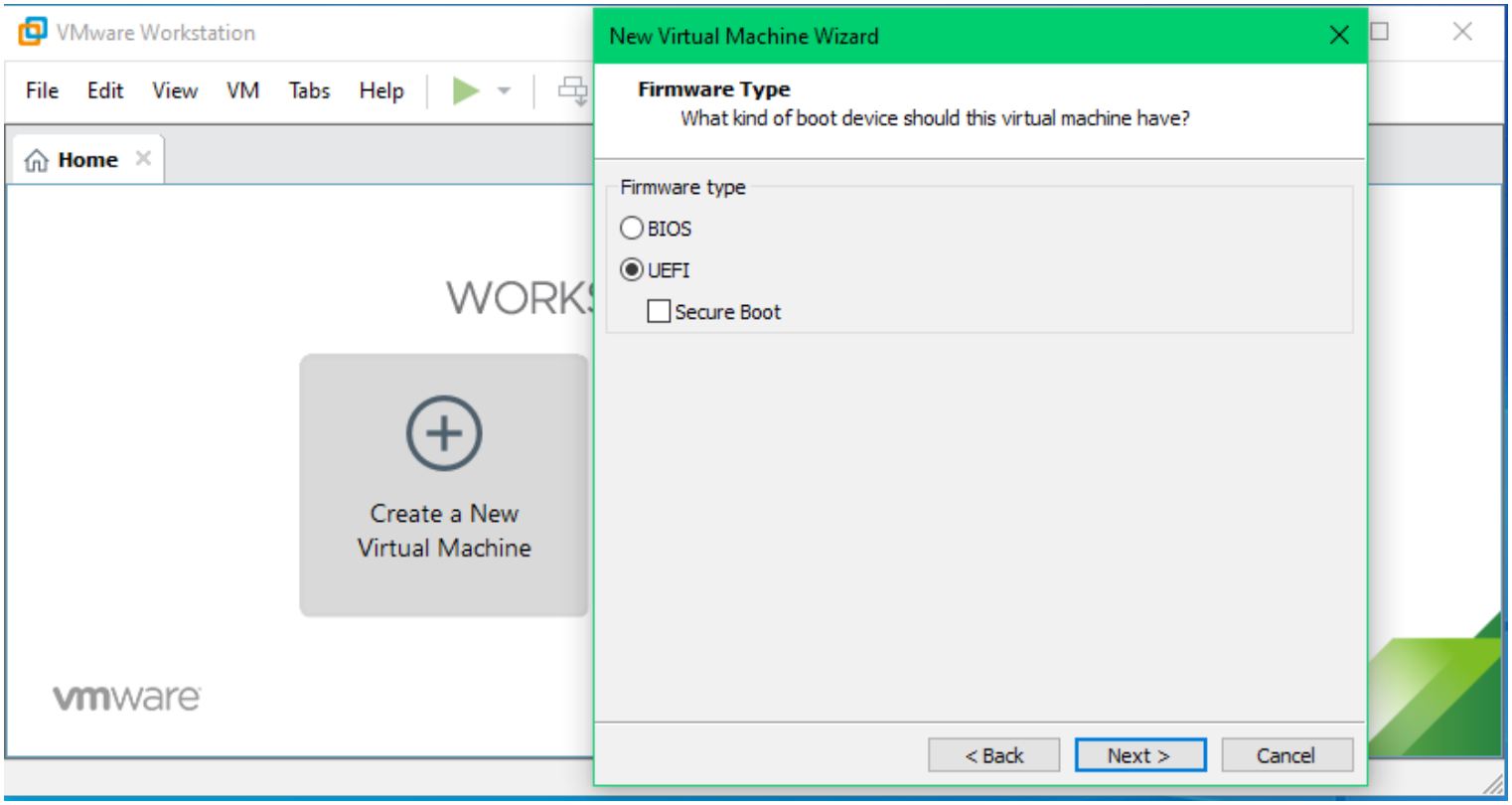
You can see in the screenshot above that I have selected the Microsoft Windows radio button and selected Windows Server 2016 from the drop-down version menu. You may be thinking “Aren’t you installing Server 2019?” well yes however, that option is not available in the drop-down list of versions. So I am simply using the next closest OS.

6. Name the Virtual Machine



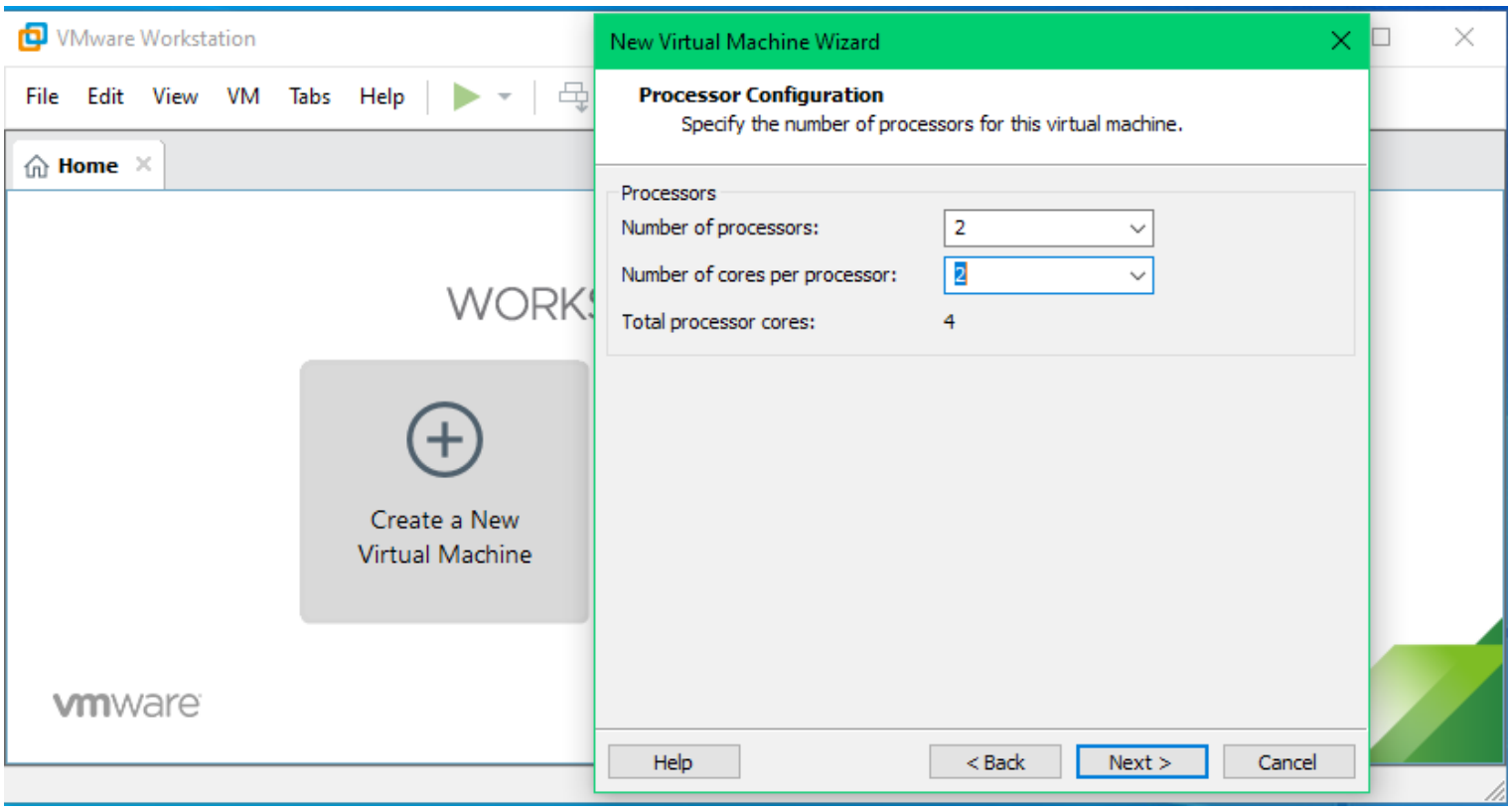
Create a meaningful name or your virtual machine, in this case I have named the VM: Server_2019_Testlab and select the location to save the Virtual machine files. I am saving it on a portable solid-state drive (SSD), mainly due to a lack of disk space on my main drive.

7. Select the Firmware Type



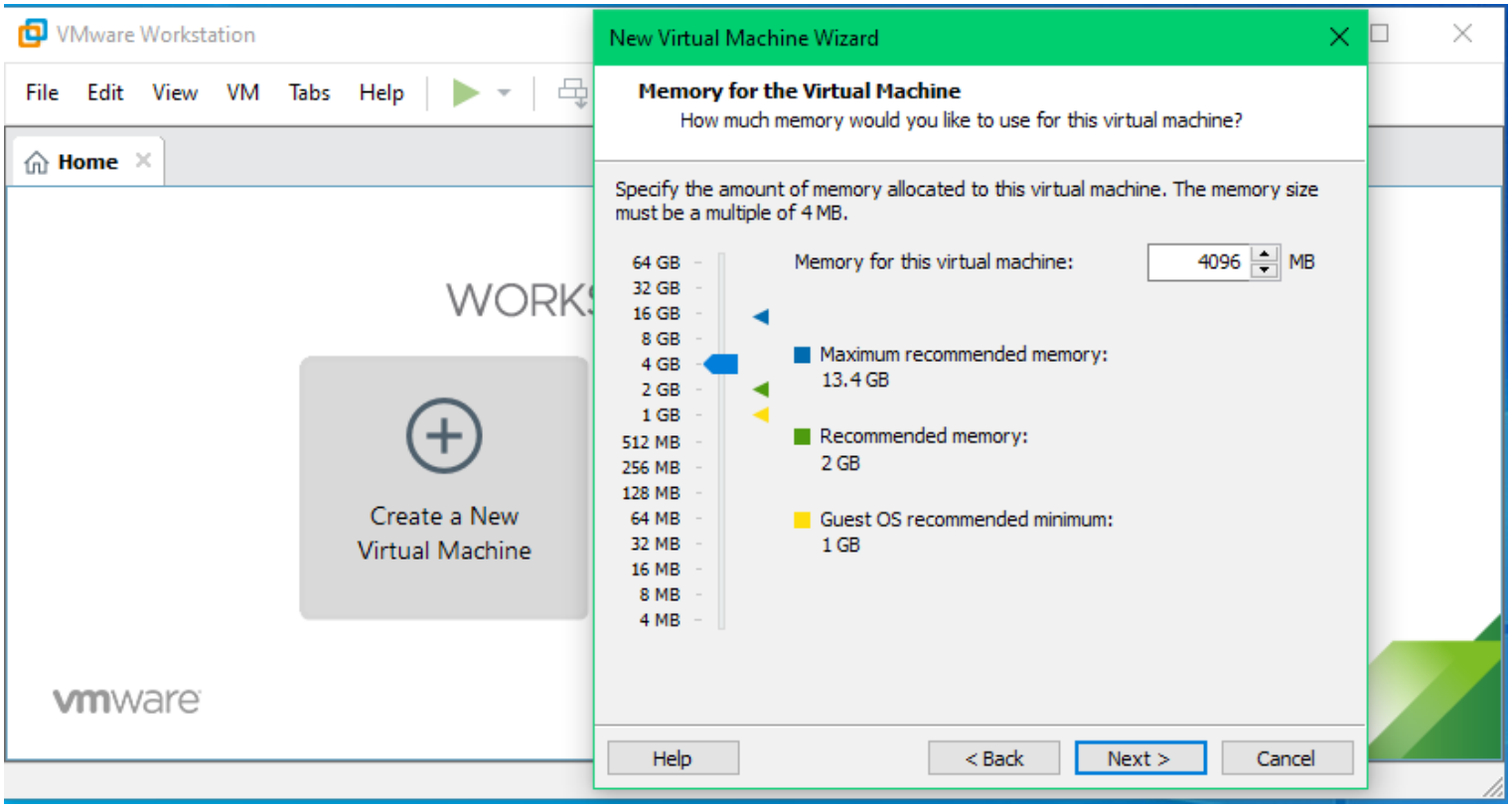
UEFI was selected as default and I will be sticking with that option.

8. Processor Configuration



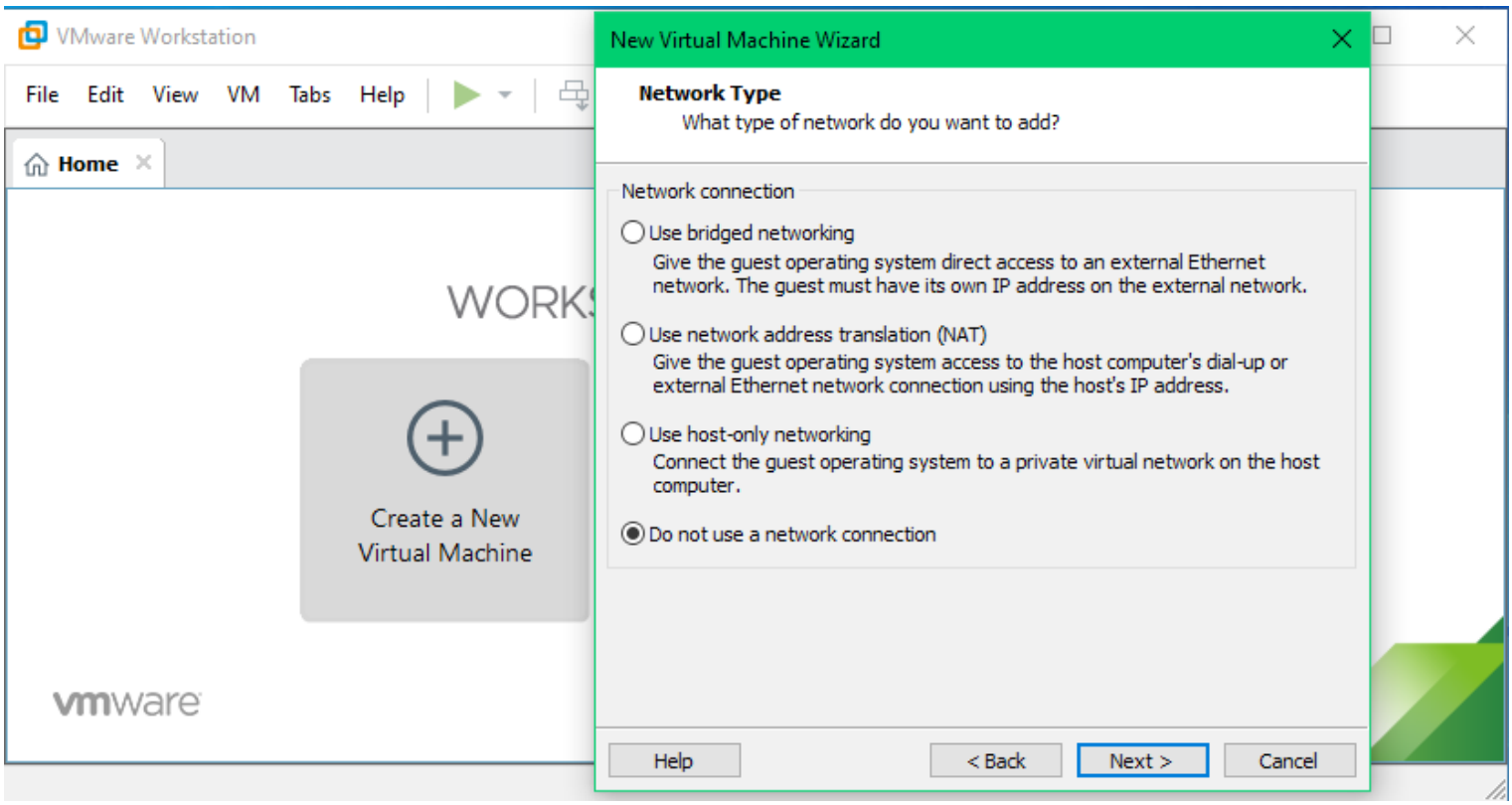
Here we are required to allocate the number of processors and processor cores or the virtual machine to use.

9. Memory Allocation (RAM)



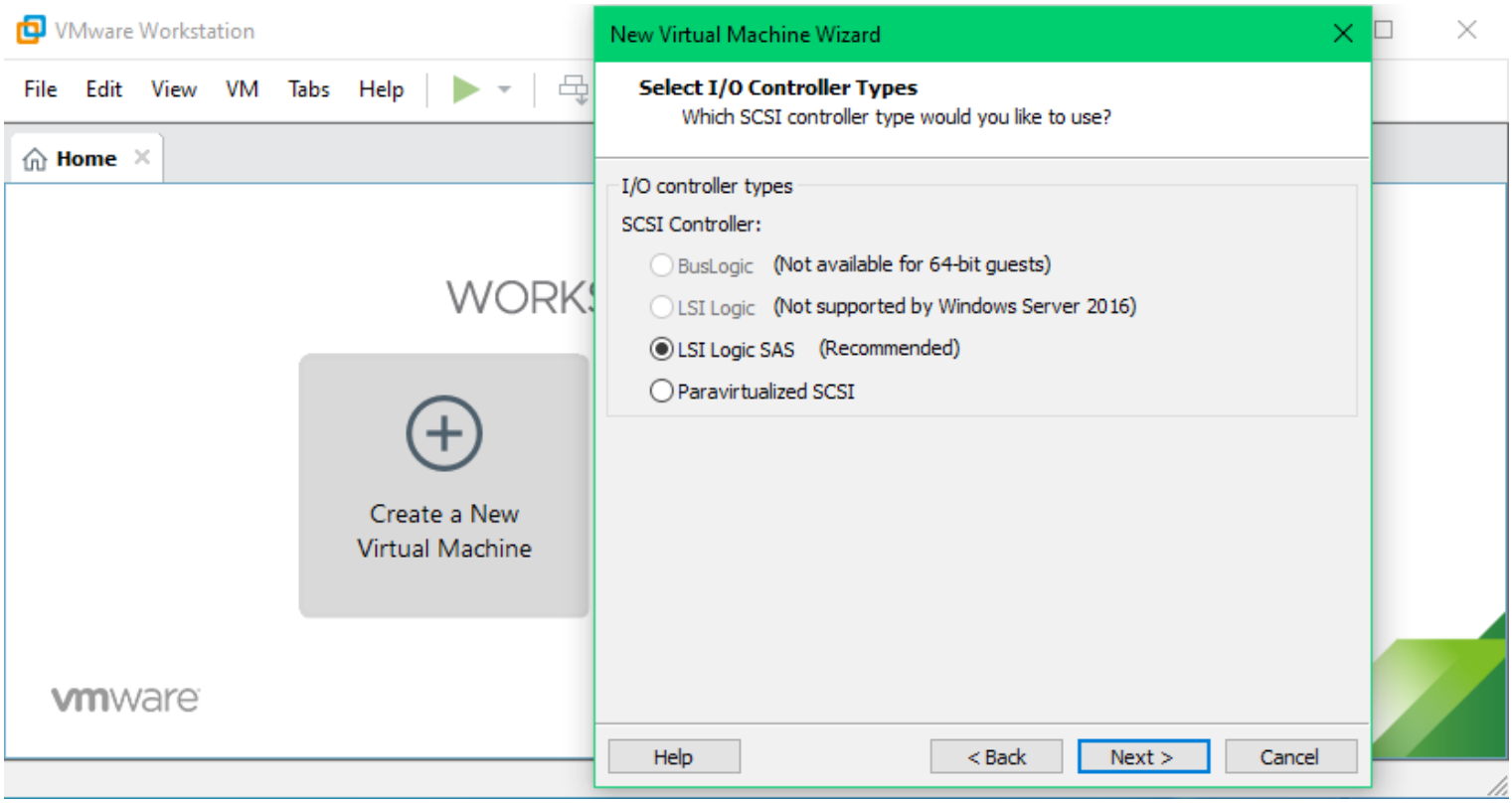
I have allocated 4GB of RAM for my virtual machine to use.

10. Network Type



I have selected the option to not use a network connection, I will configure this prior to powering on the virtual machine or the first-time.

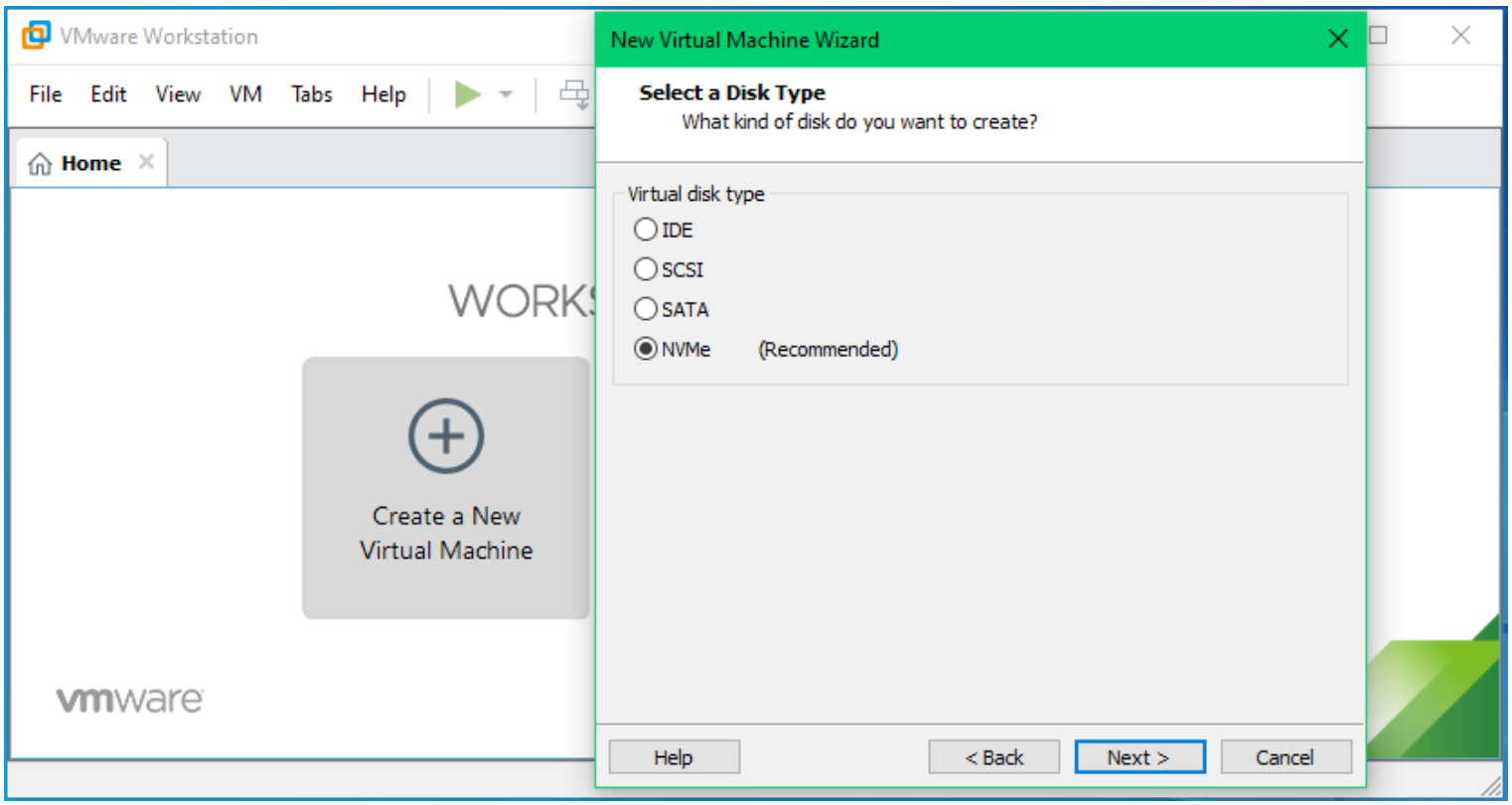
11. SCSI I/O Controller Type



For the SCSI I/O controller I have selected the default option of LSI Logic SAS.

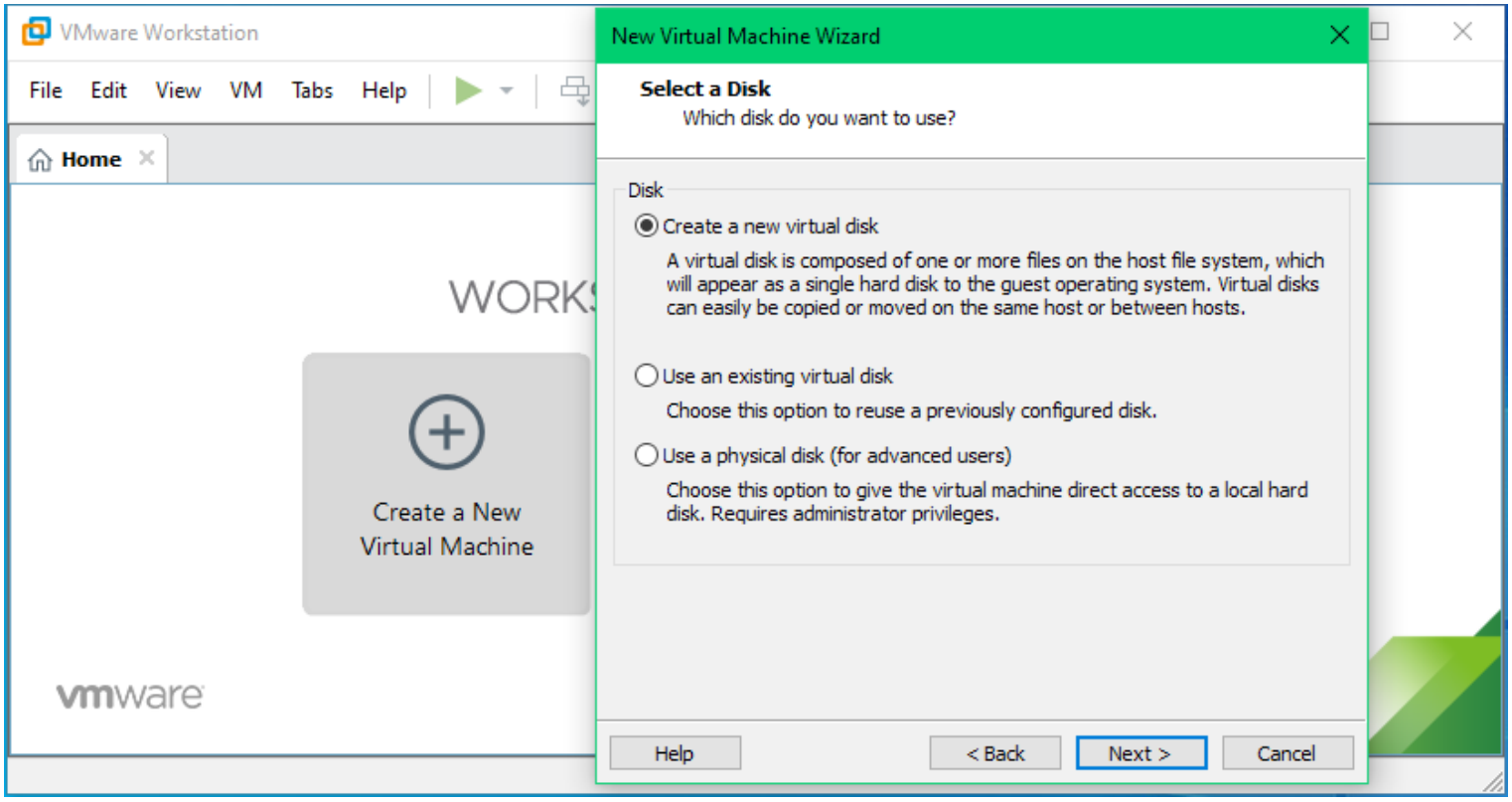
SCSI I/O Controllers are used by virtual machines to access other SCSI devices such as; hard-disks and CD/DVD drives.

12. Select a Hard-disk Type



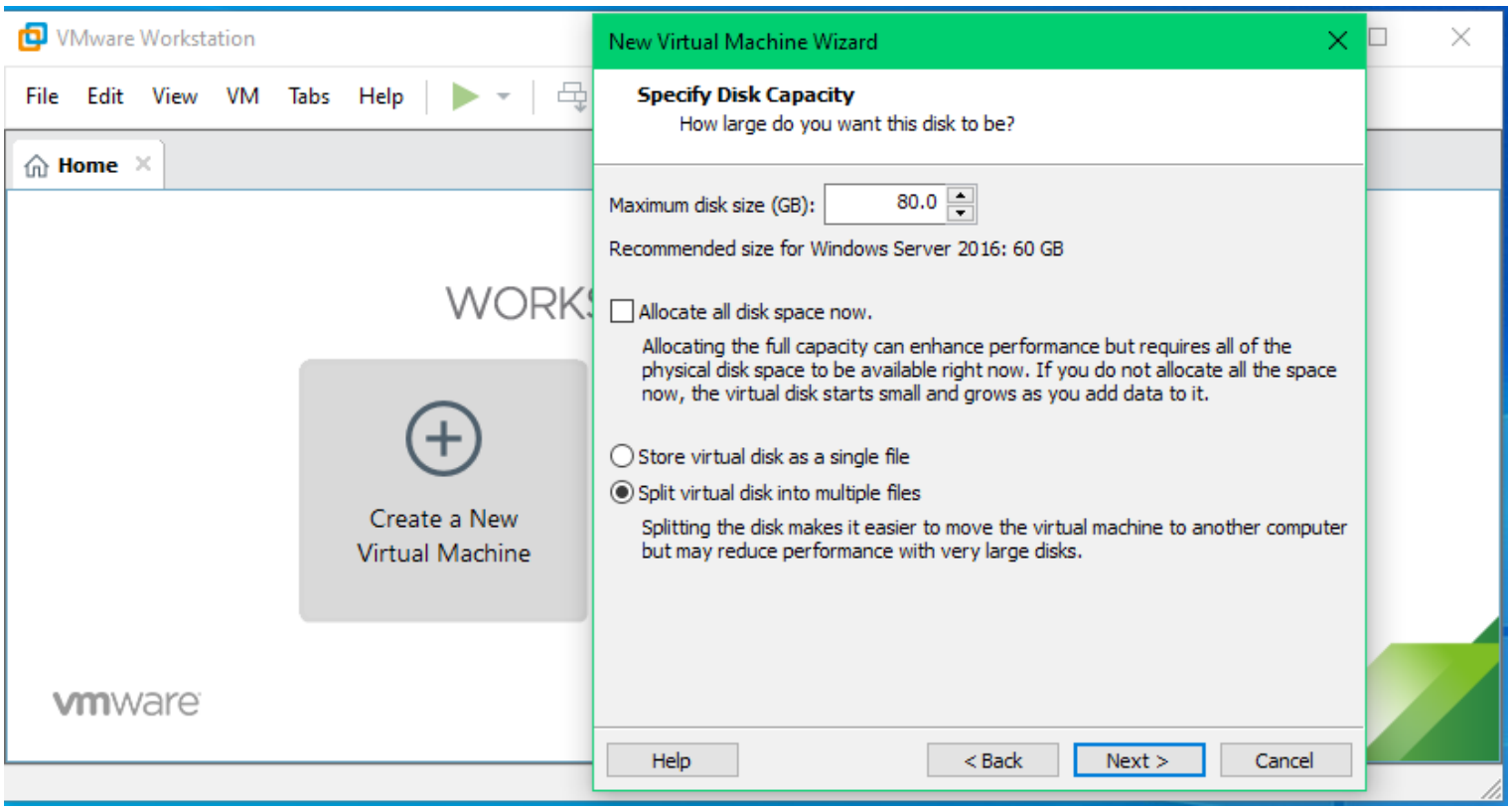
For the hard-disk type I have selected NVMe, it is the most modern disk-type available and also the recommended type for VM setup.

13. Select a Disk



I have selected the option to create a new virtual disk, as this is a brand new virtual machine no other options are applicable.

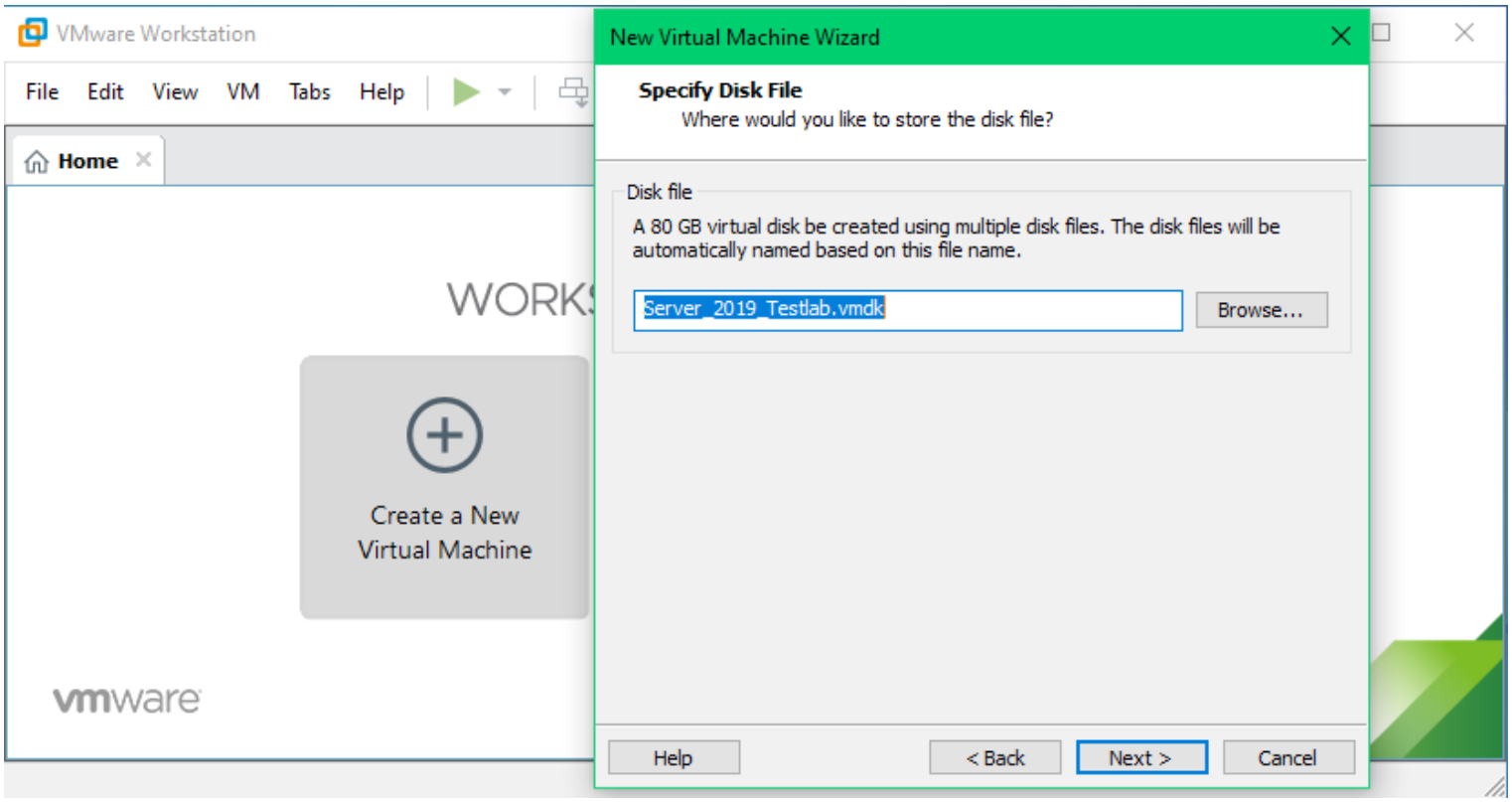
14. Specify the Capacity of the Hard-disk



I have specified for the hard-disk to be a maximum size of 80GB and for the hard-disk file to be split into multiple files.

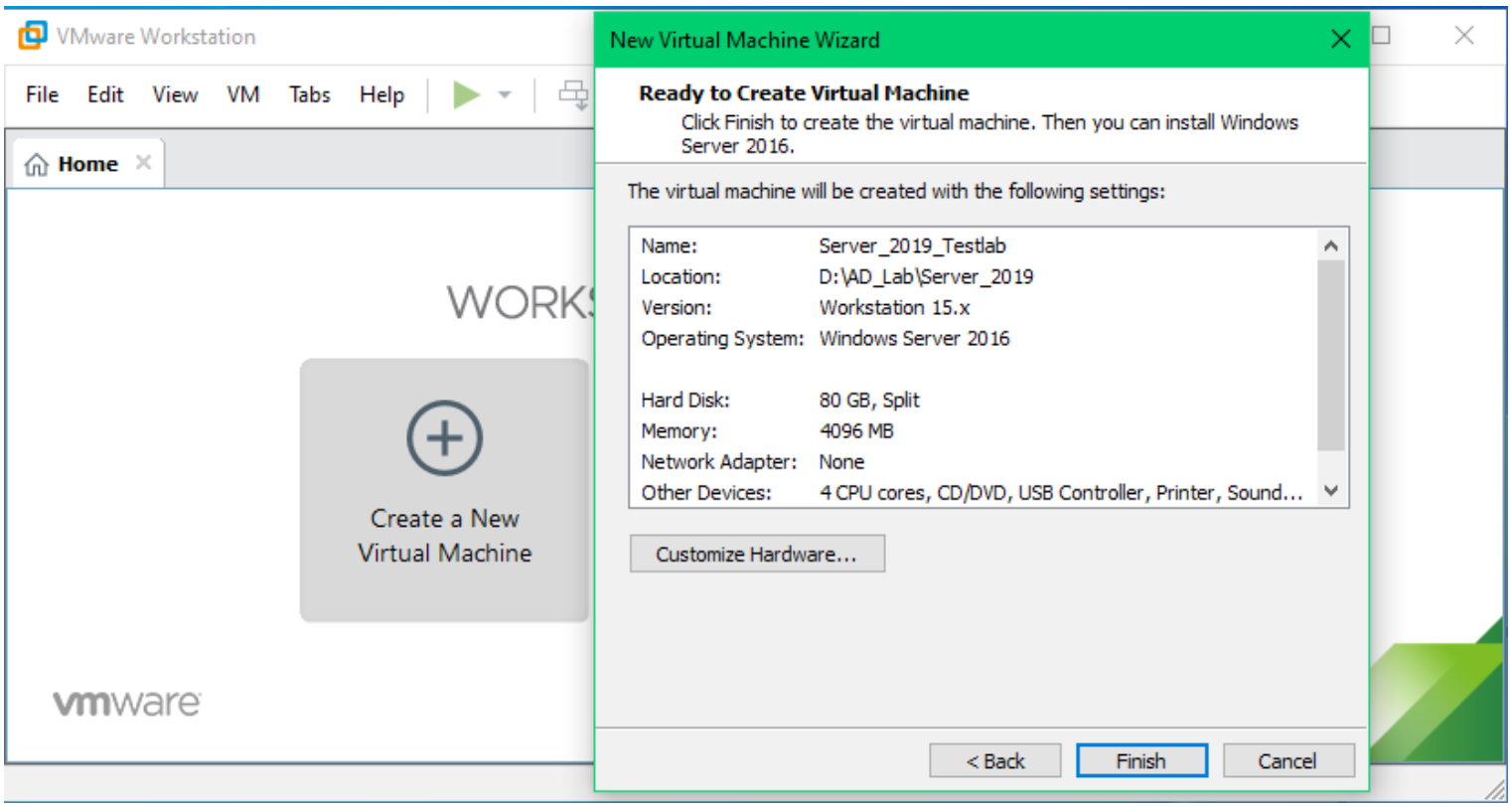
Note: The virtual-disk image will start at around 512kb before expanding once we install the guest OS and additional services. The 80GB file-size will not be reached for quite some time (possibly never).

15. Specify the Disk's Filename



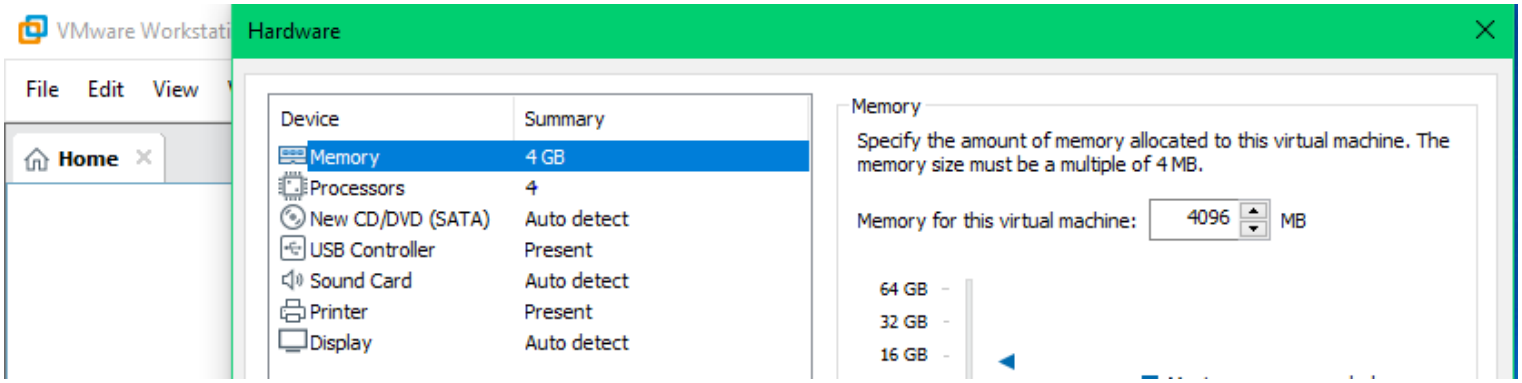
I have simply allowed the disk file-naming convention to remain the default option. If you wish to name your disk something else, here is the place to do it.

16. Not Quite – Ready to Create Virtual machine

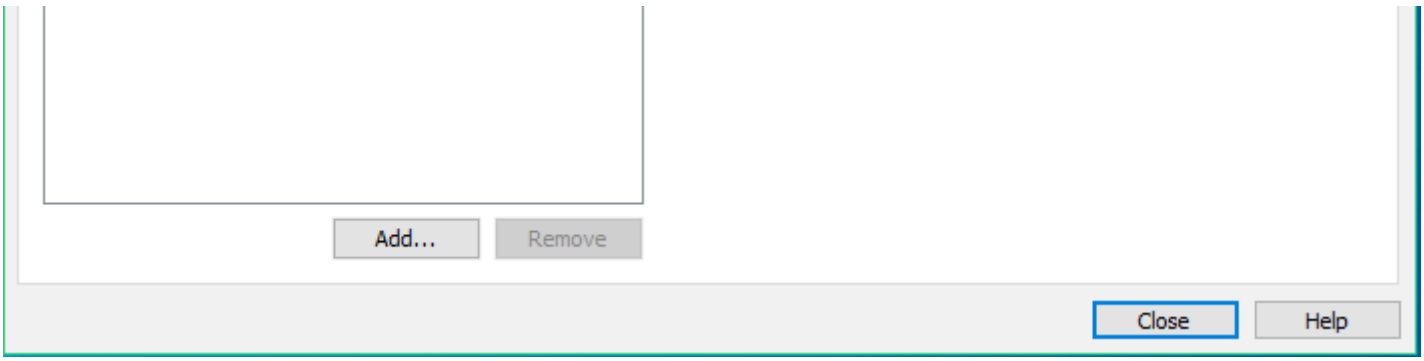


Here we are presented with an overview of the setting we have configured so far throughout the virtual machine creation wizard. Whilst we were working through the wizard we skipped past the OS and Network adaptor options, to configure these click the customise hardware button.

17. Additional Hardware Configuration

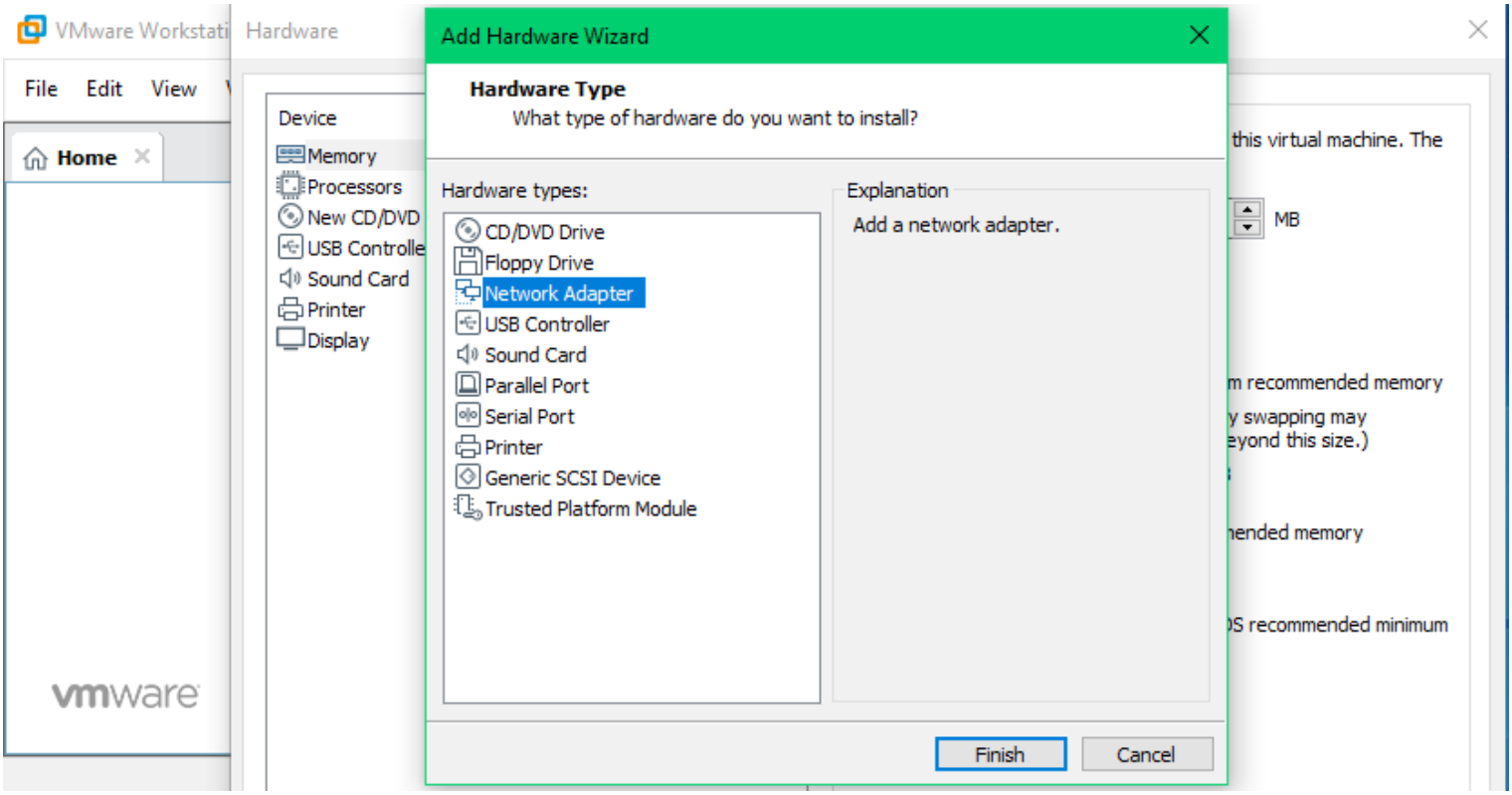


You can see we now have a more detailed view and the ability to add and or reconfigure hardware.



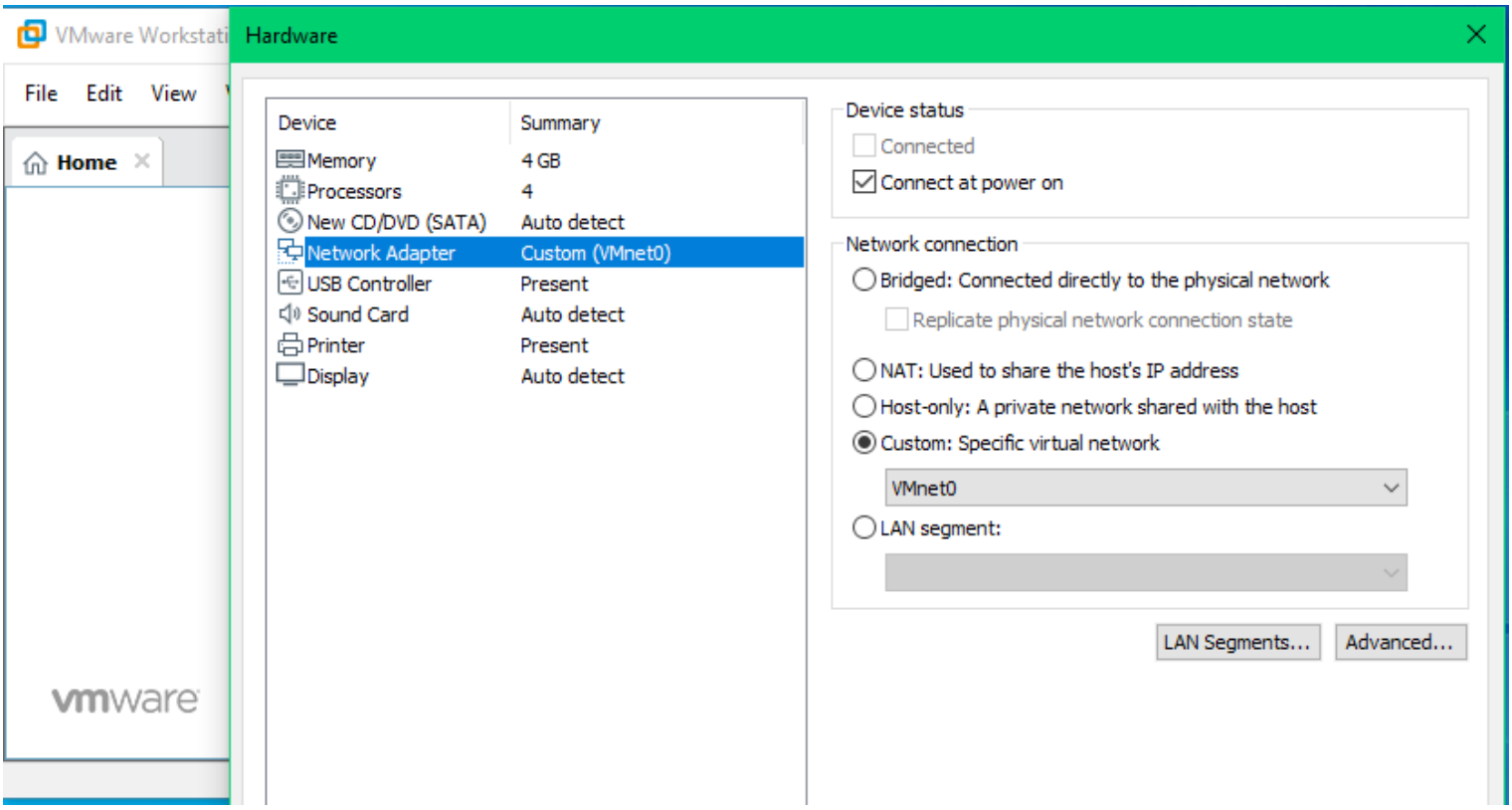
By Clicking the Add… button it will launch the add hardware wizard.

18. Add Hardware Wizard



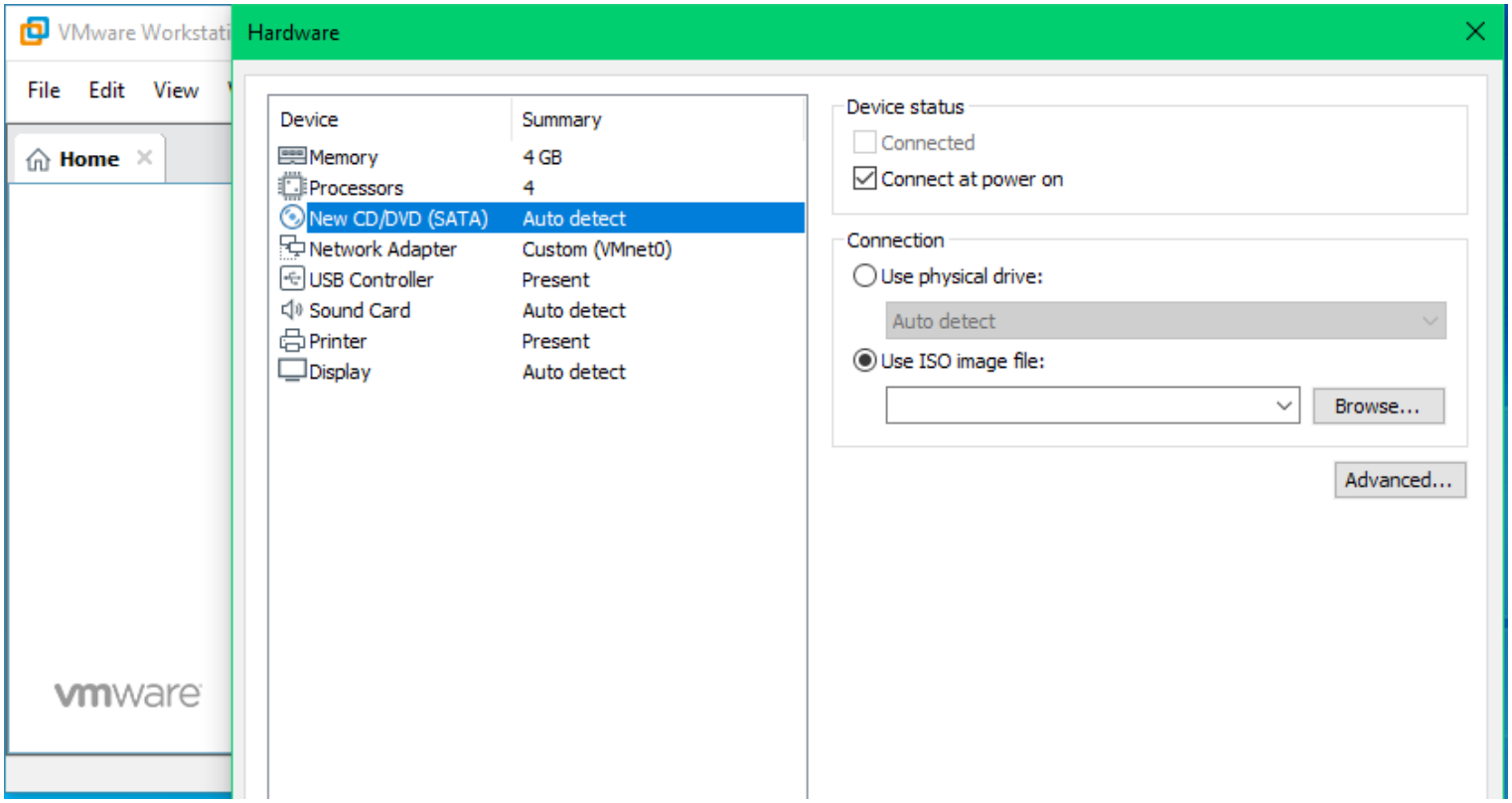
From the add hardware wizard I have selected Network Adaptor and then proceeded to click finish.

19. Network Adaptor



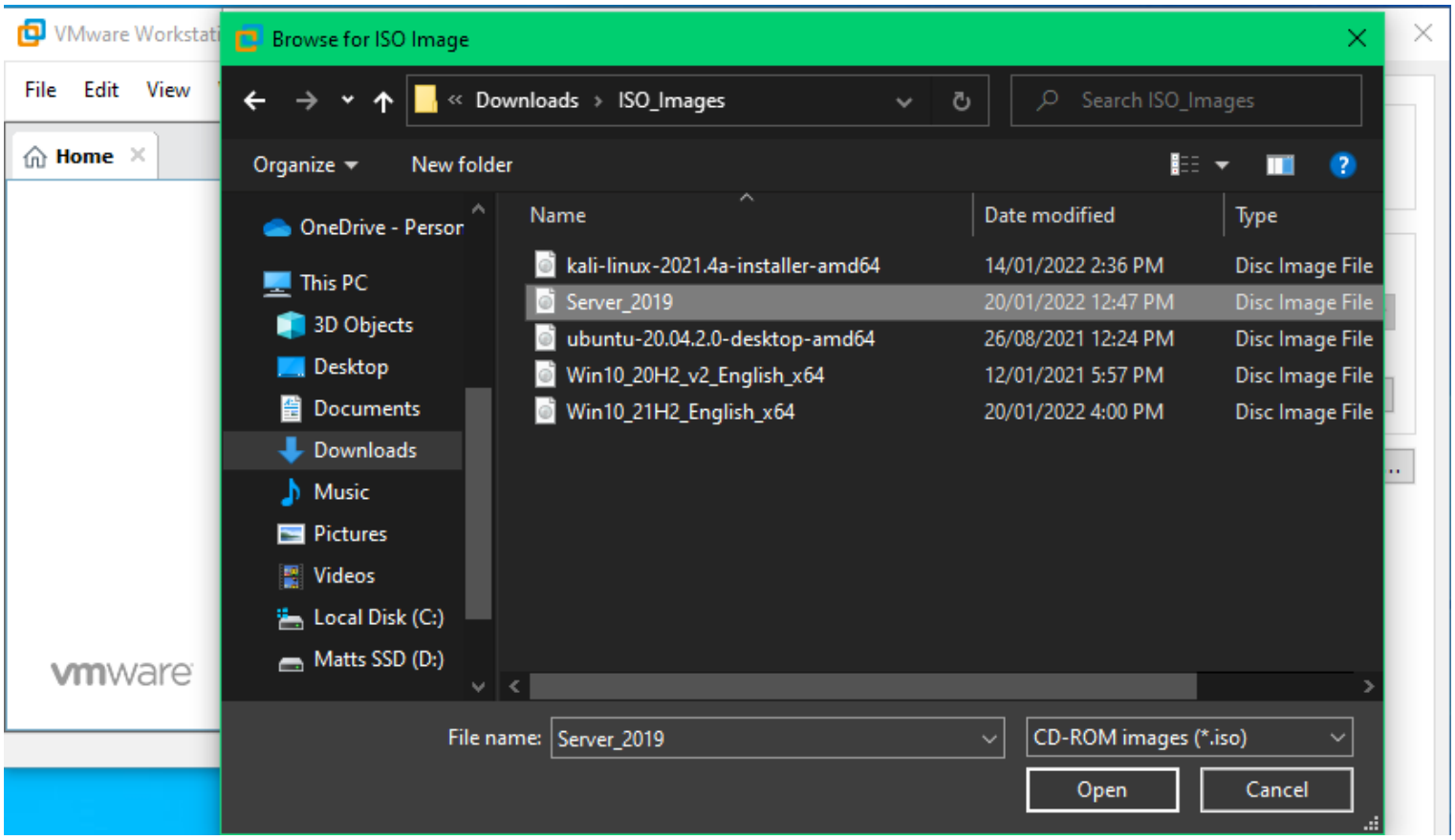
This is the configuration settings of the network adaptor that we just added to our virtual machine. You can see in the screenshot I have selected the custom radio button and then selected VMnet0 from the drop-down box. On my host I have configured VMnet0 as directly bridged to my laptops wireless card, which allows connection to my local wireless network. The virtual machine will be assigned an IP address from my networks DHCP server when the device is powered on.

20. CD/DVD Configuration (For our OS image file)



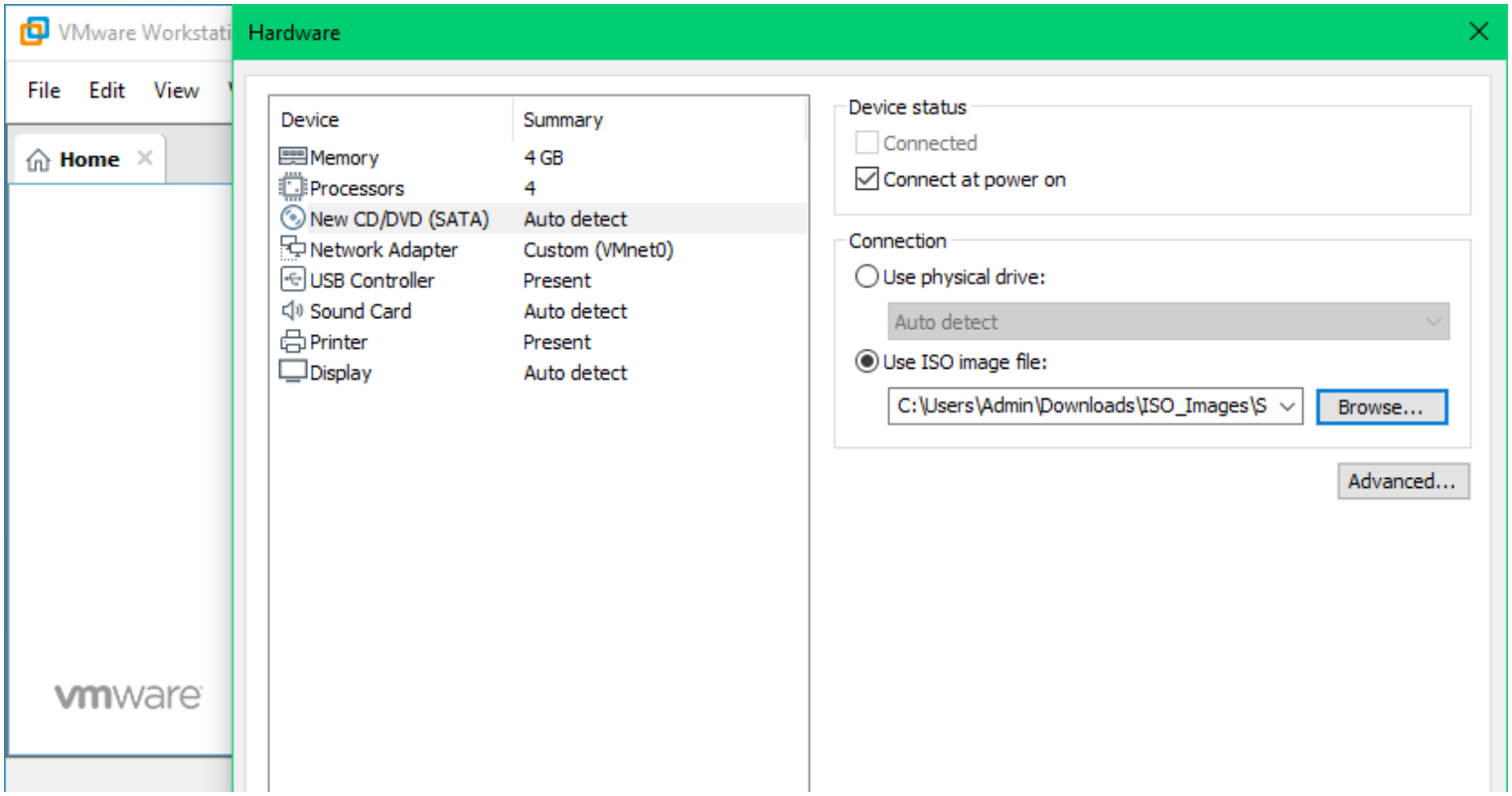
I have now selected New CD/DVD before selecting use ISO image file and clicking browse on the right hand side of the hardware pane. This is where we will point the CD drive at our Guest OS's ISO file, which will then be mounted at boot.

21. Navigate to the Guest OS ISO File.



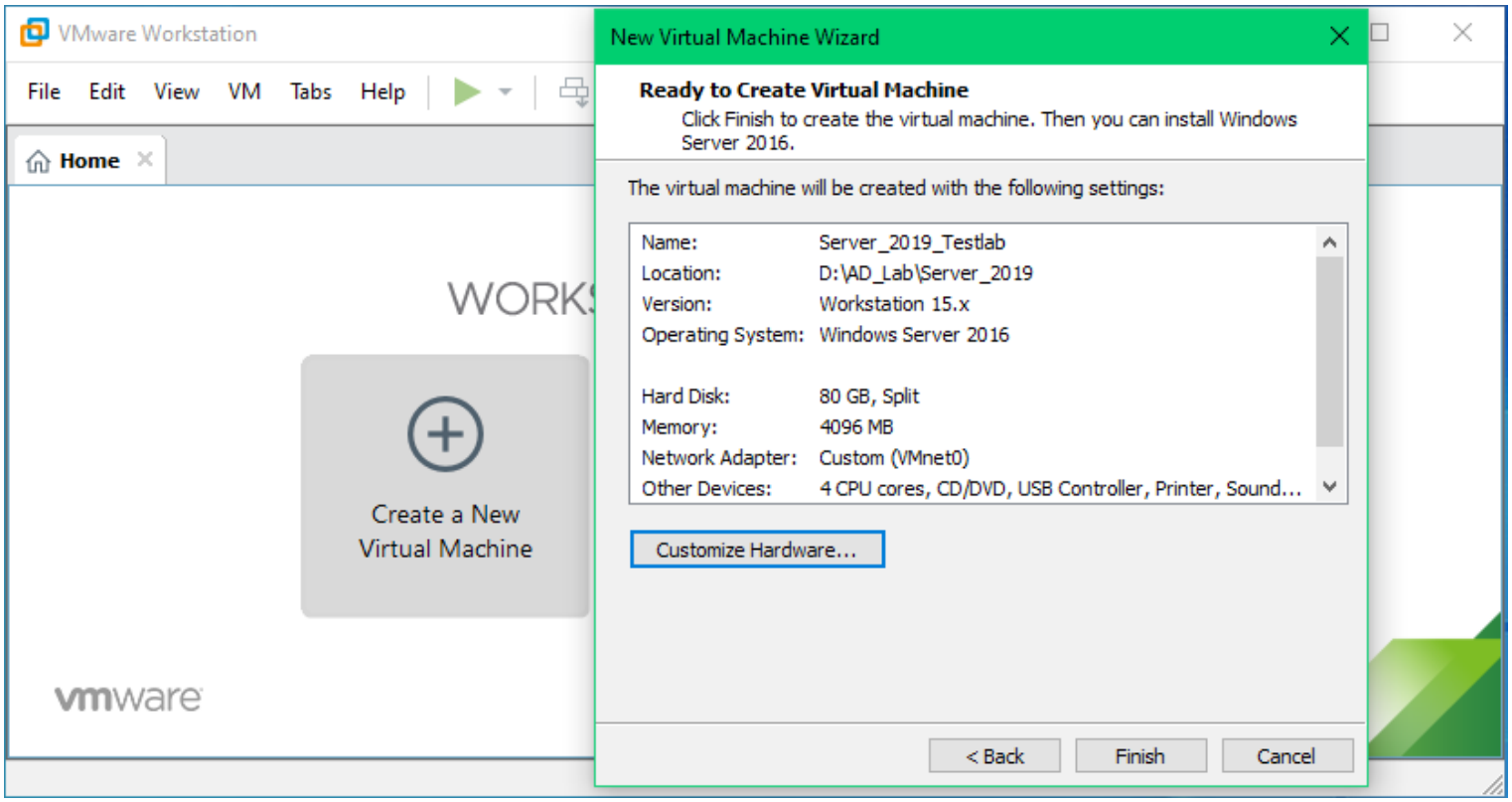
You can see I have navigated to the directory where I store my OS ISO files, I have selected Server_2019 and will now simply click open.

22. Hardware Configuration



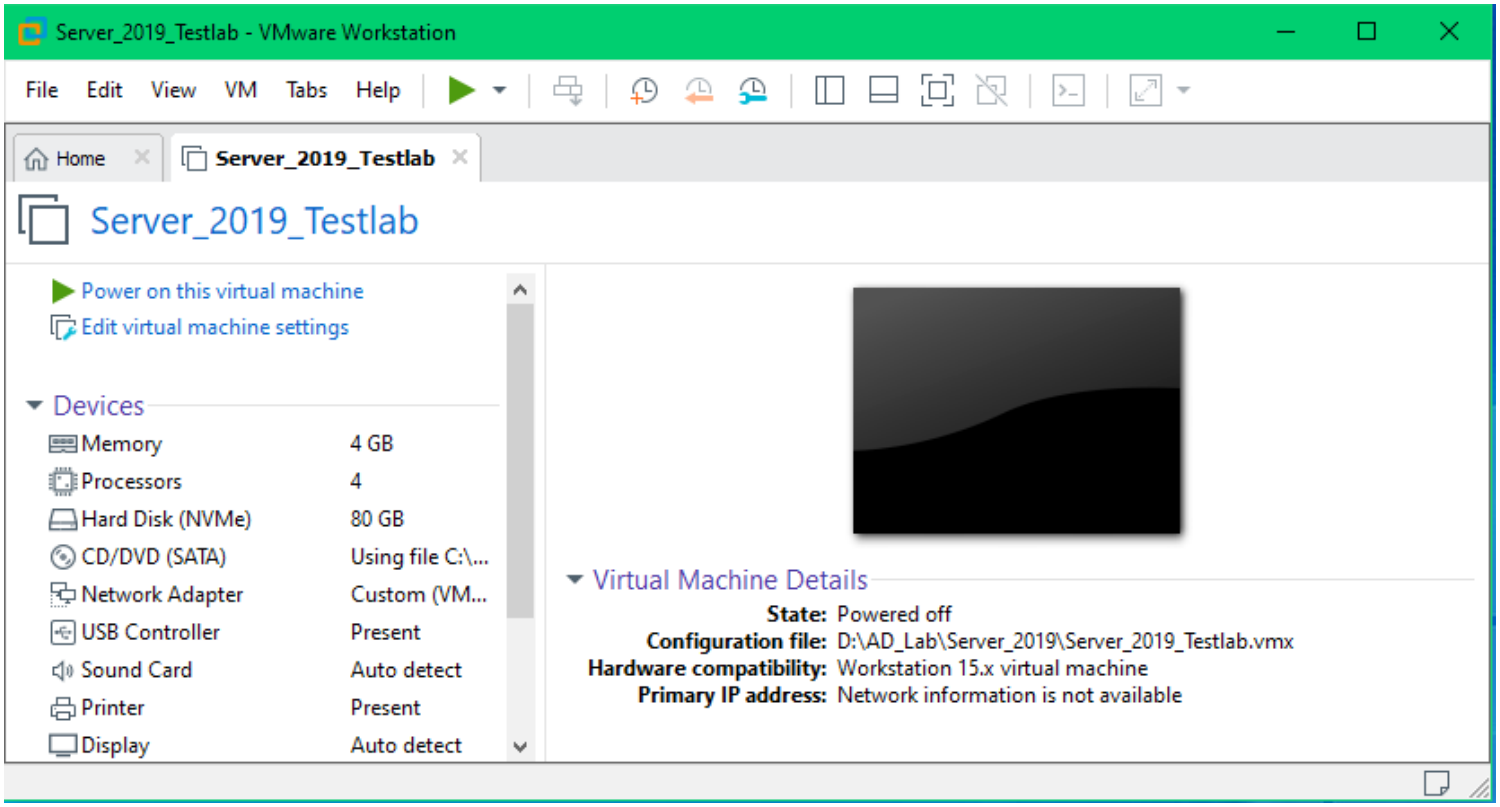
You can now see that the path to the guest OS's ISO file is populated, I will now exit the hardware configuration window by pressing close at the bottom of the window.

23. Ready to Create Virtual Machine



We have now finished the virtual machine creation wizard, you can now simply click finish and this will commence the creation of the virtual machine.

24. Ready to Boot



The Windows Server 2019 virtual machine is ready to boot and install the guest OS.

Lets just quickly recap on what we have configured so far;

- Selected a guest operating system - When asked by the setup wizard which guest OS we would be installing in the virtual machine, I selected Microsoft Windows Server 2016 as Server 2019 in not available in the drop-down menu.

- Specified the amount of physical hardware to allocate – When asked by the wizard how much memory and the number of CPU cores to allocate, I specified 4096 MB or 4 GB of RAM and 2 Processors each with 2 cores.

- Specified the type of disk or the virtual machine to use – When prompted by the wizard to specify a disk type, I went with the recommended option of NVMe. When asked which disk we would like to use, I specified to create a new virtual disk-file.

- Set maximum disk-size and how to store the virtual disk-files – The wizard asks for the maximum desired size of the disk, to which I specified 80 GB. I also selected the option to split the virtual-disk into multiple files.

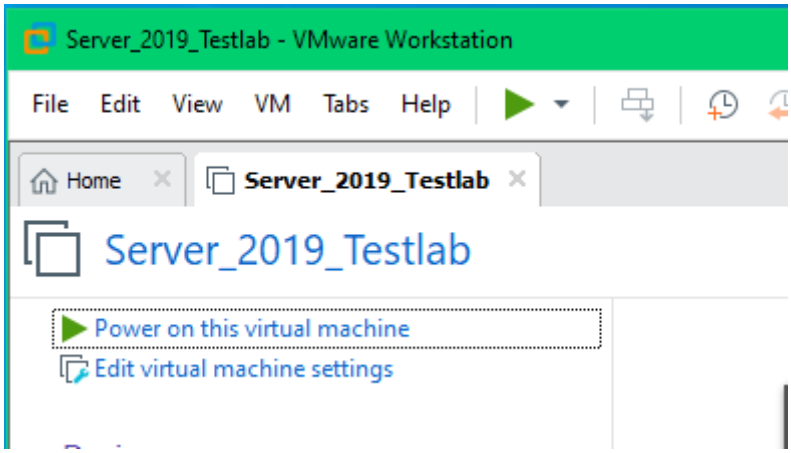
- Added a network adaptor – When prompted by the wizard to add a network adaptor, I originally chose not to use a network connection. Prior to exiting the creation wizard I customised the hardware and added a network adaptor which is bridged directly to my laptops wireless card.

- Configured the guest OS ISO to be mounted via the virtual CD drive at first boot – When the wizard asked what OS we would like to install I selected install the OS later. Prior to finishing the wizard I entered the customise hardware menu and configured the virtual CD/DVD drive to load our ISO file at first boot.

- Created virtual machine – Upon completing the customisation of the hardware, I clicked finish which then began creating the VM. Within seconds I have a virtual machine ready for our OS of choice to be installed.

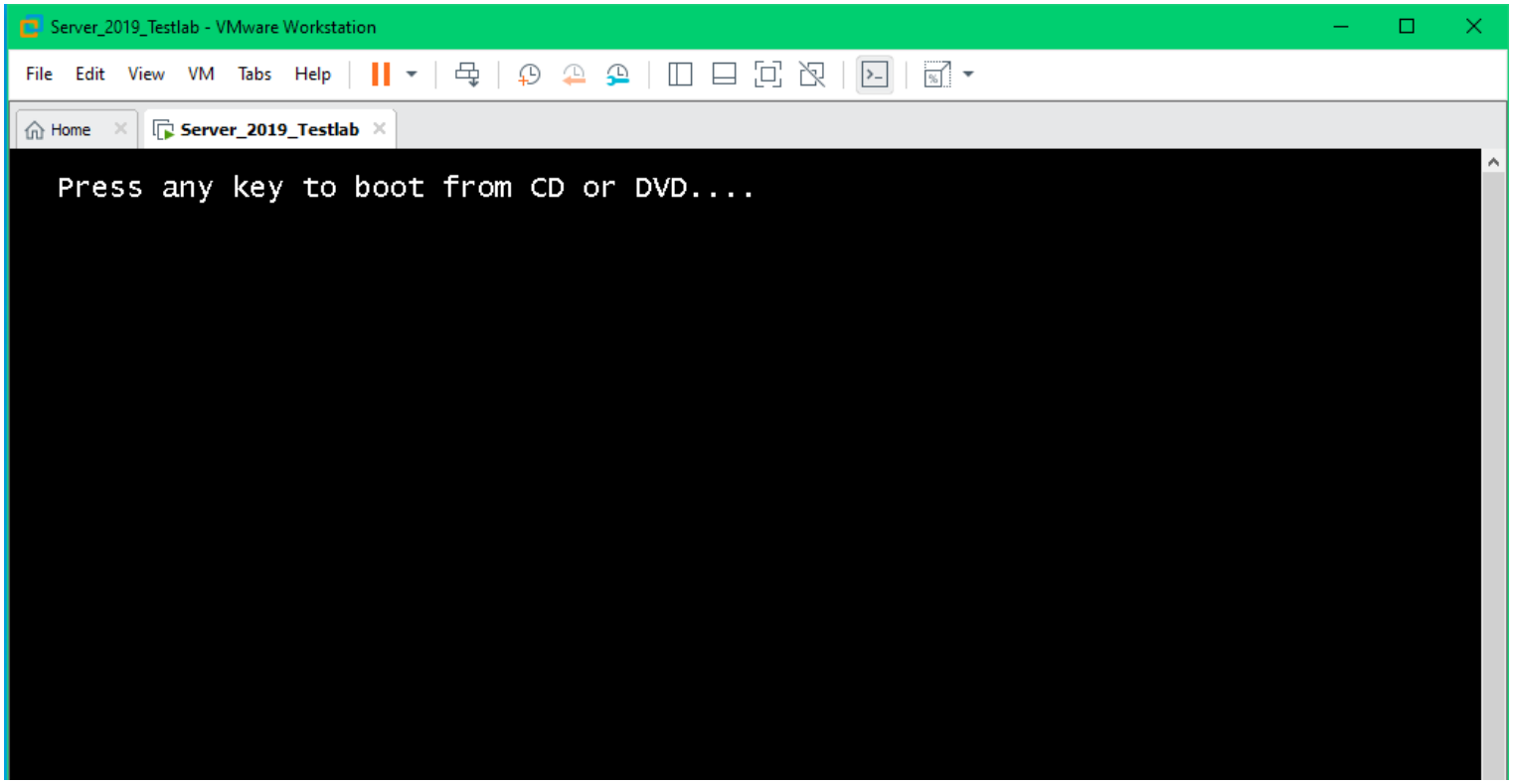
Windows Server 2019 Install - Guest OS Install

1. Power on the Virtual Machine



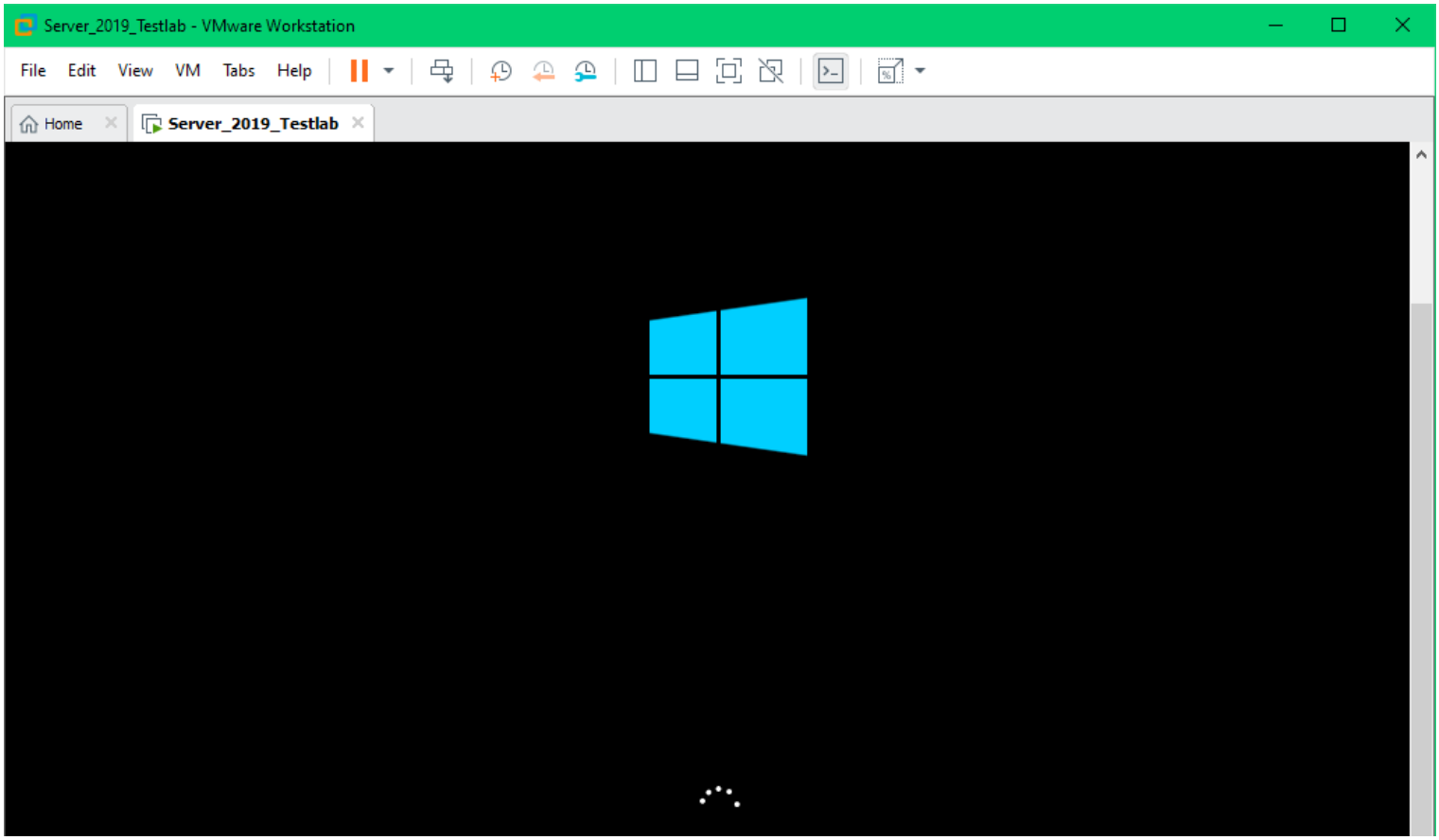
By clicking Power on this virtual machine the VM will begin the boot process or in our case being first-boot the ISO image in the virtual-disk drive will be mounted to the virtual machine.

2. Boot From CD/DVD Drive



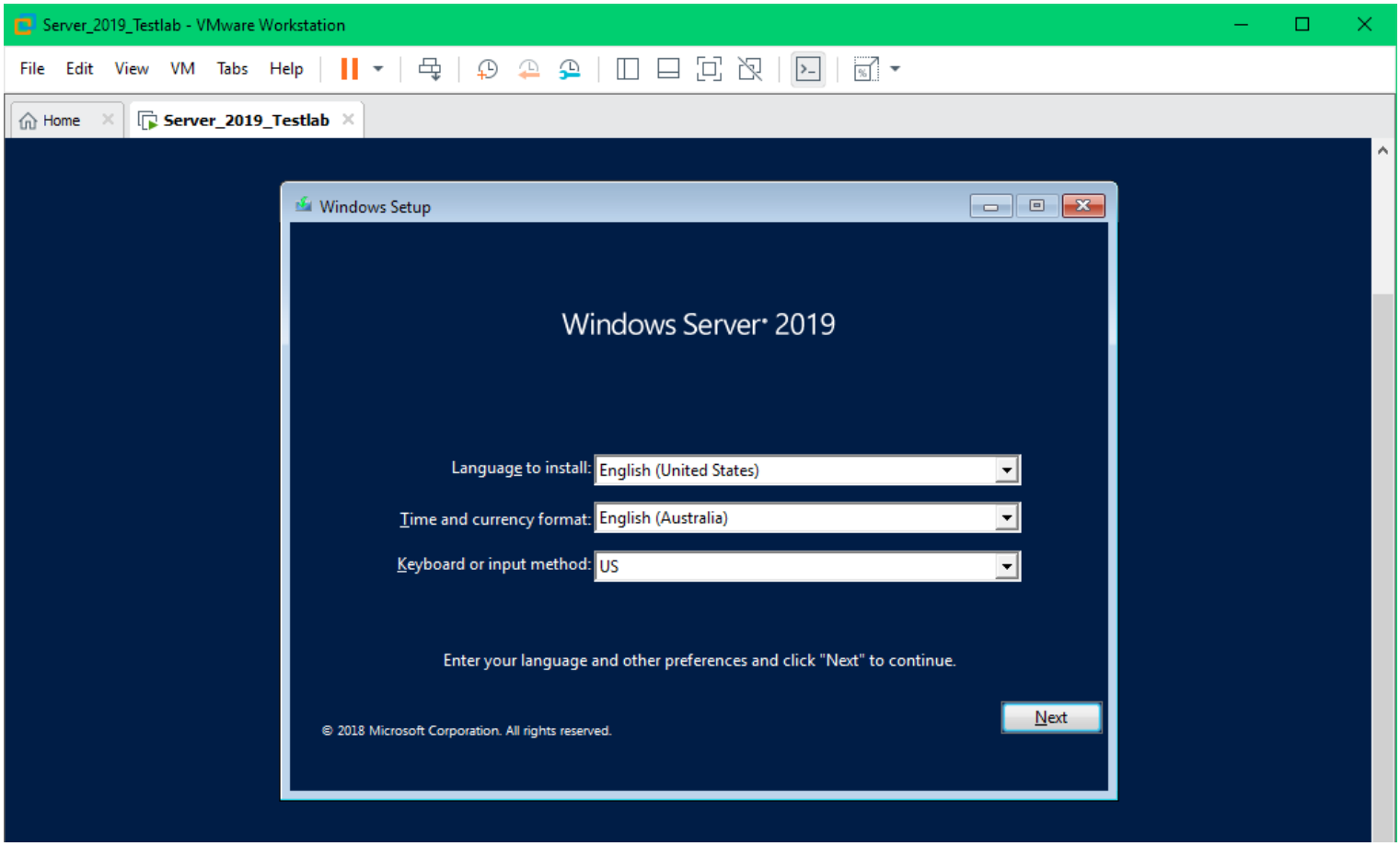
This is the splash screen presented by the Windows Server ISO image upon boot, you have about 5 seconds to click within the guest OS's screen area and press absolutely any key (I prefer the space bar).

3. Windows Getting Things Ready



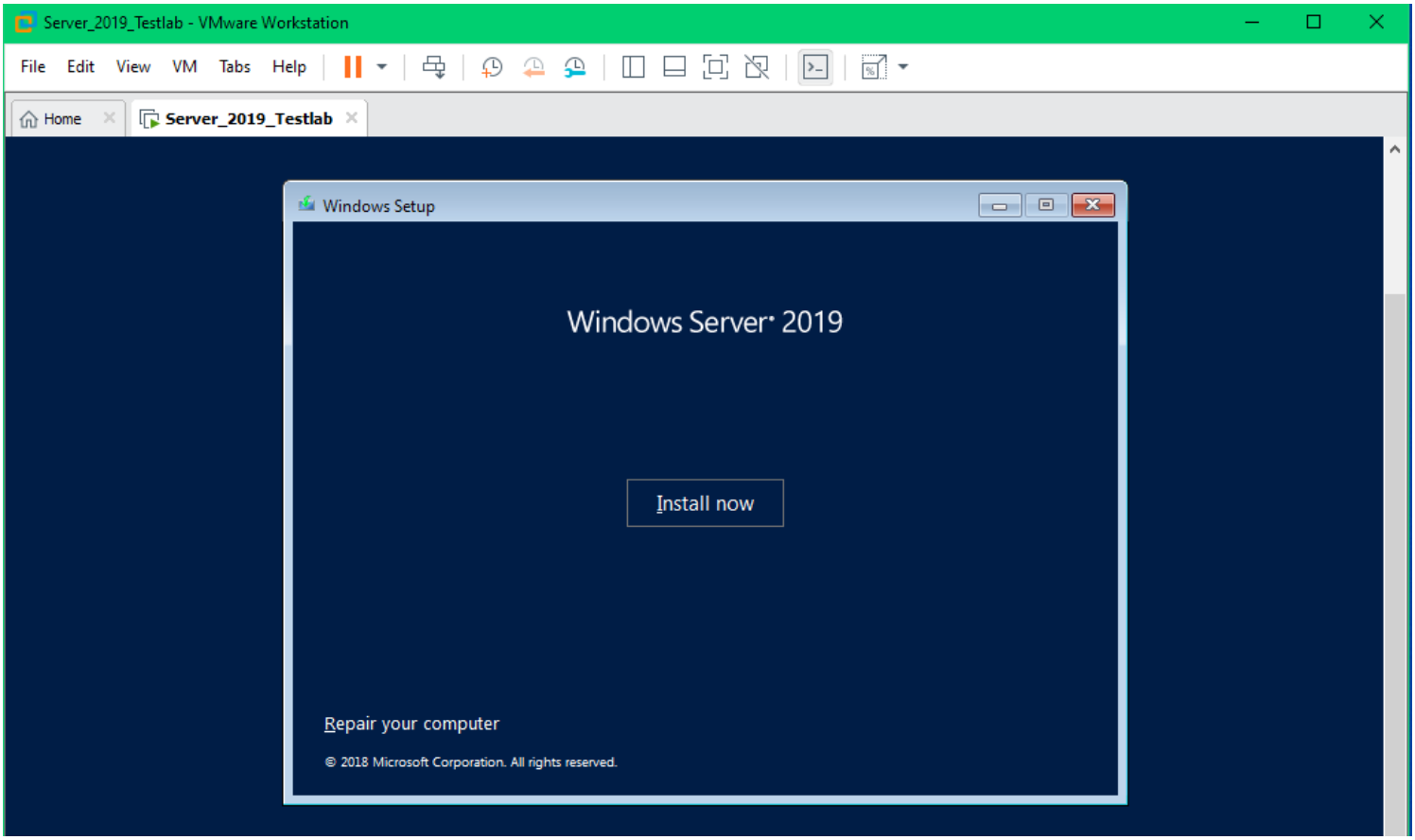
As Windows installers usually do, its just getting some things ready before we can proceed.

4. Windows Setup – Initial Options



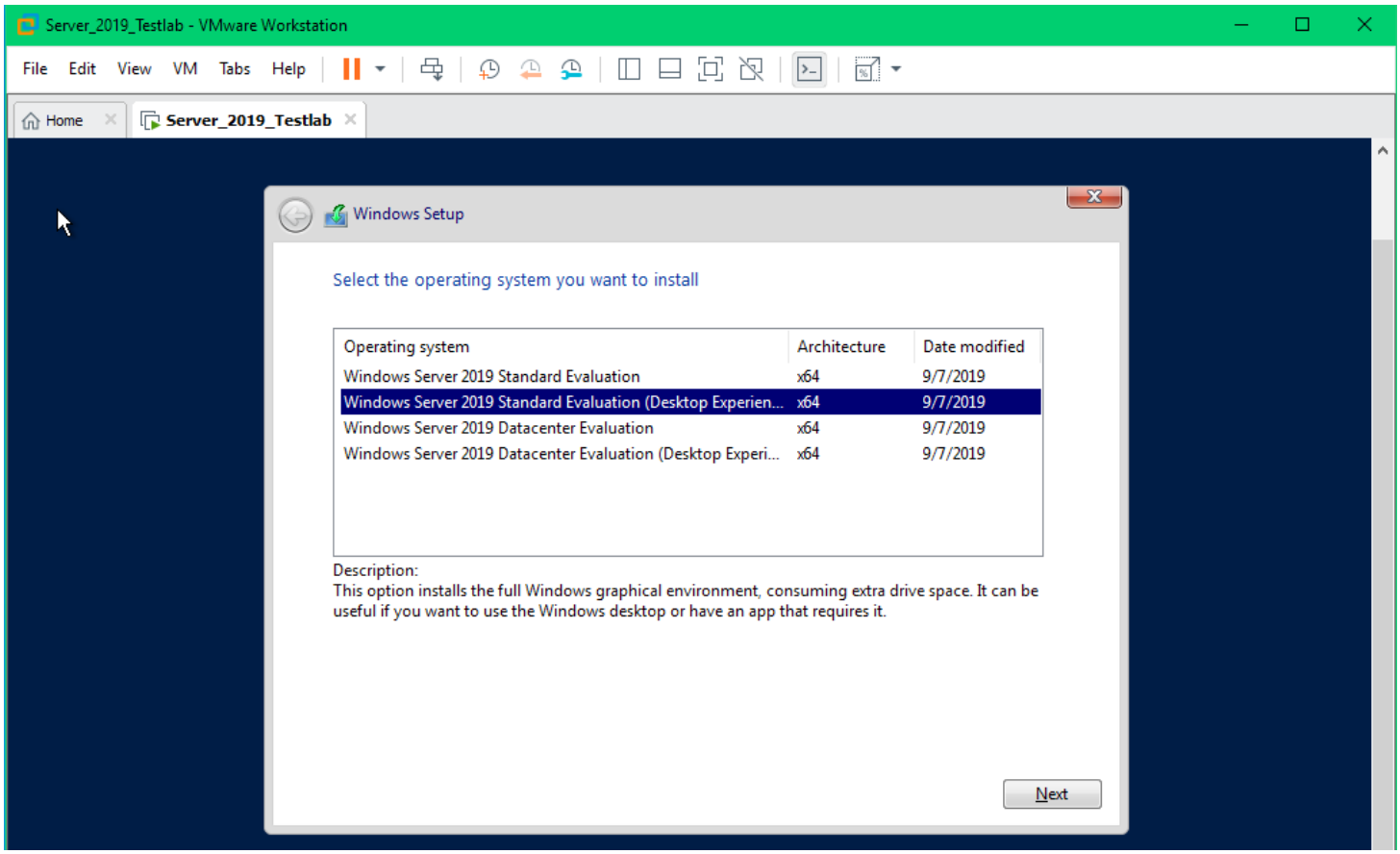
Select your Language to install, time & currency format and keyboard or input method. Then press next!

5. Windows Setup - Begin Installation Process



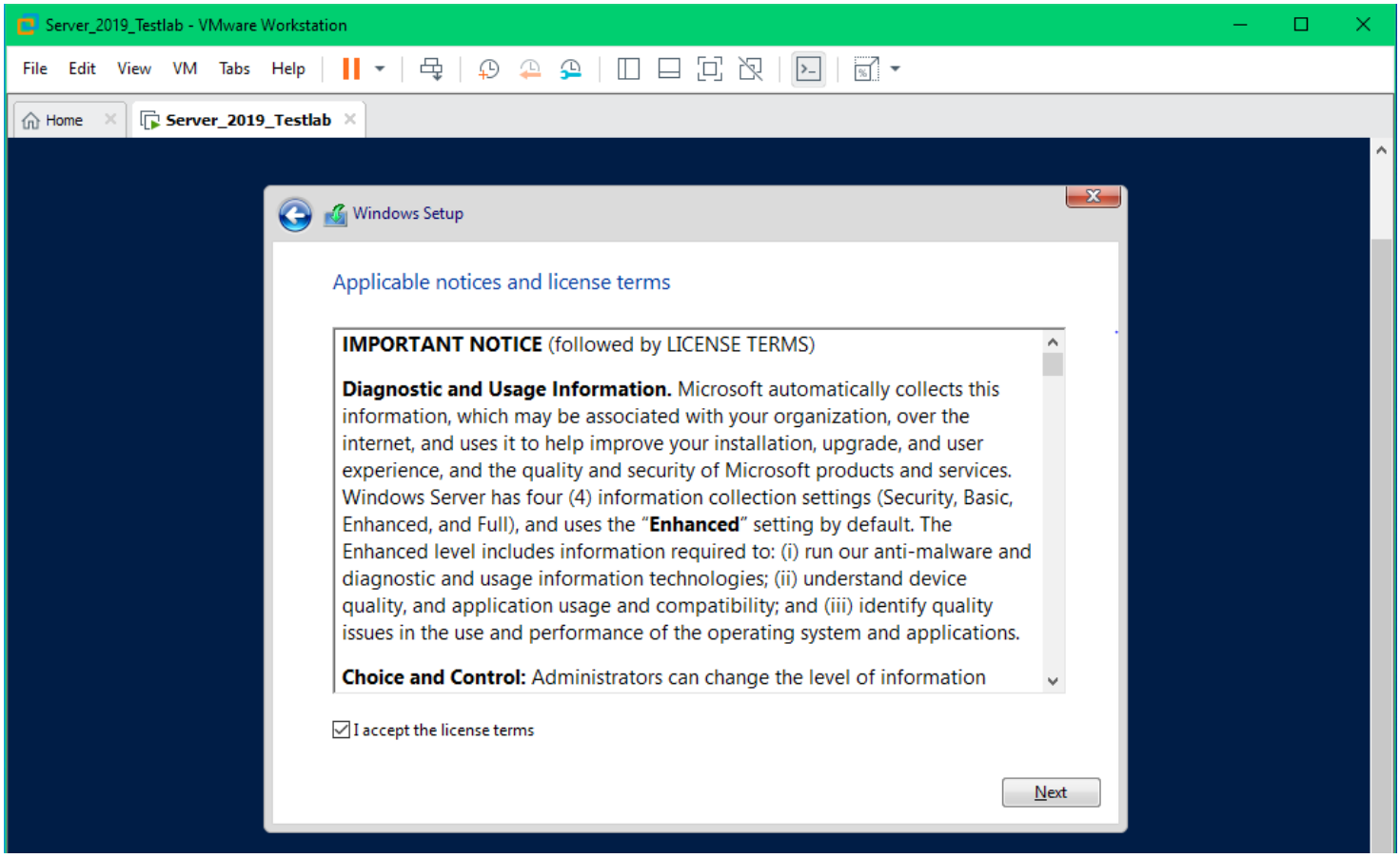
By clicking the Install now button this will begin the Windows Setup wizard to install the guest OS.

6. Windows Setup – Select the OS to Install



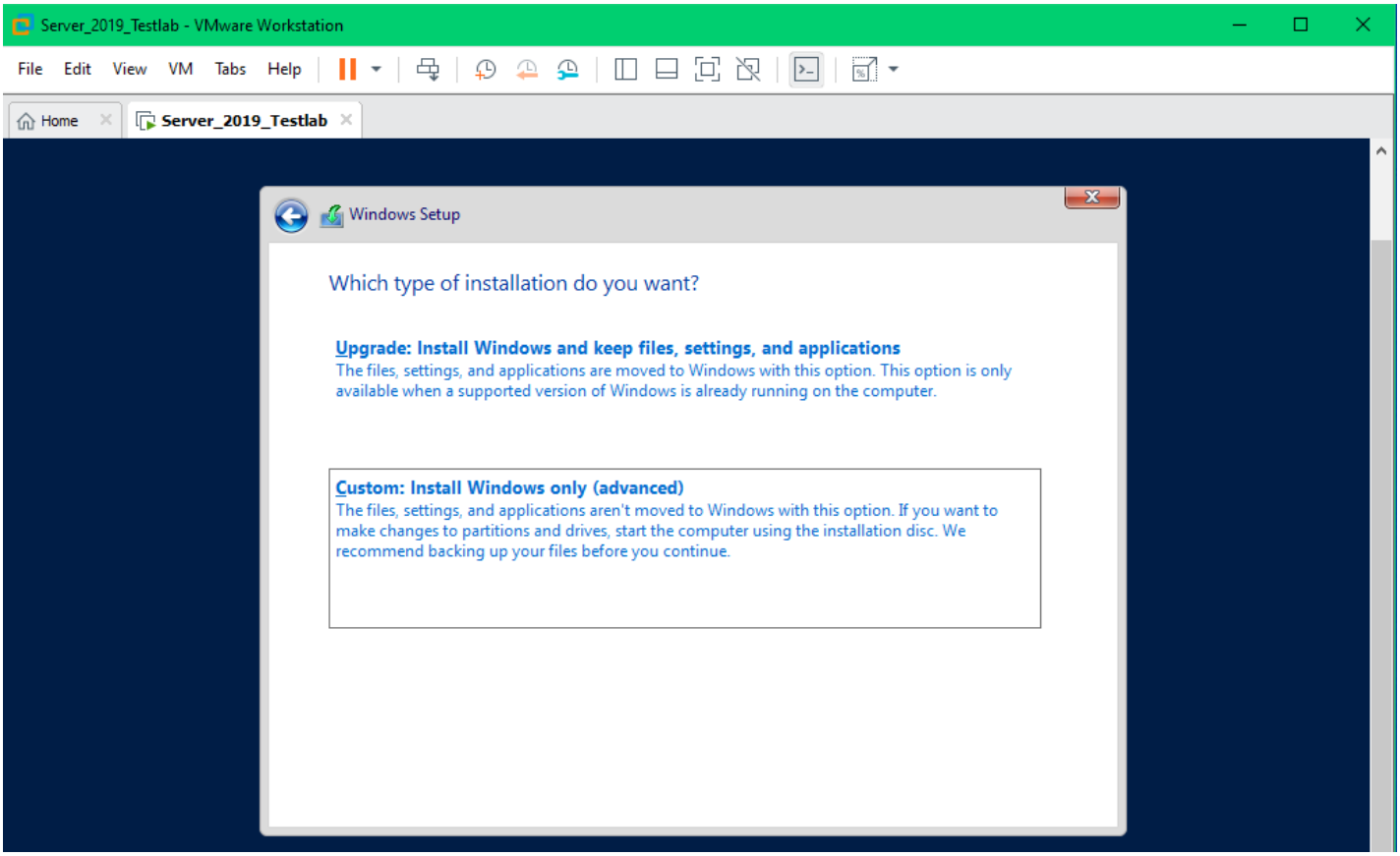
Select the particular OS Version you would like to install, for this tutorial I am installing Windows Server 2019 with a graphical user interface (GUI), known by Microsoft as the Desktop Experience.

7. Accept the Applicable Notices and License Terms



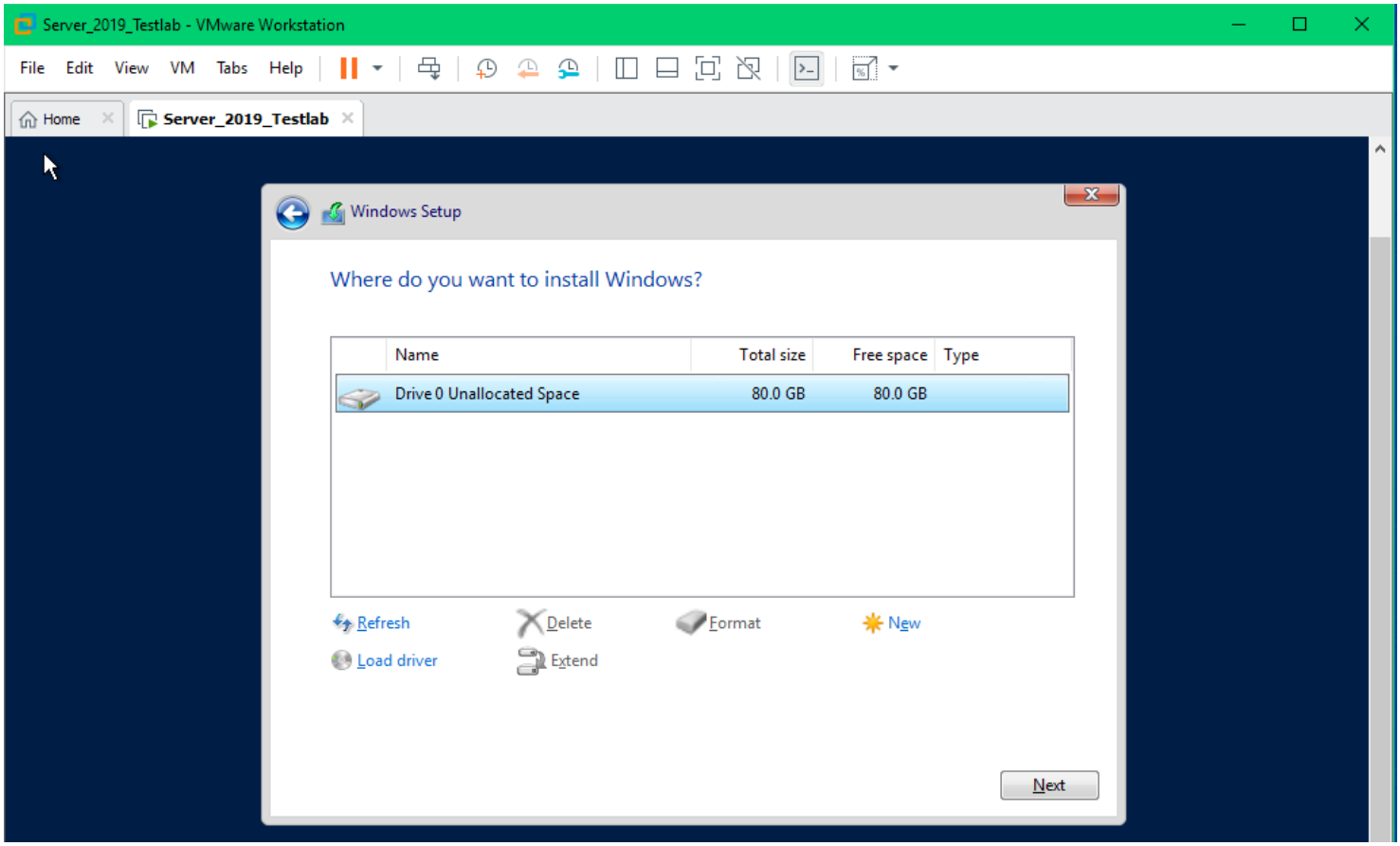
To proceed with the installation you must accept the license terms and click next.

8. Select the Installation Type



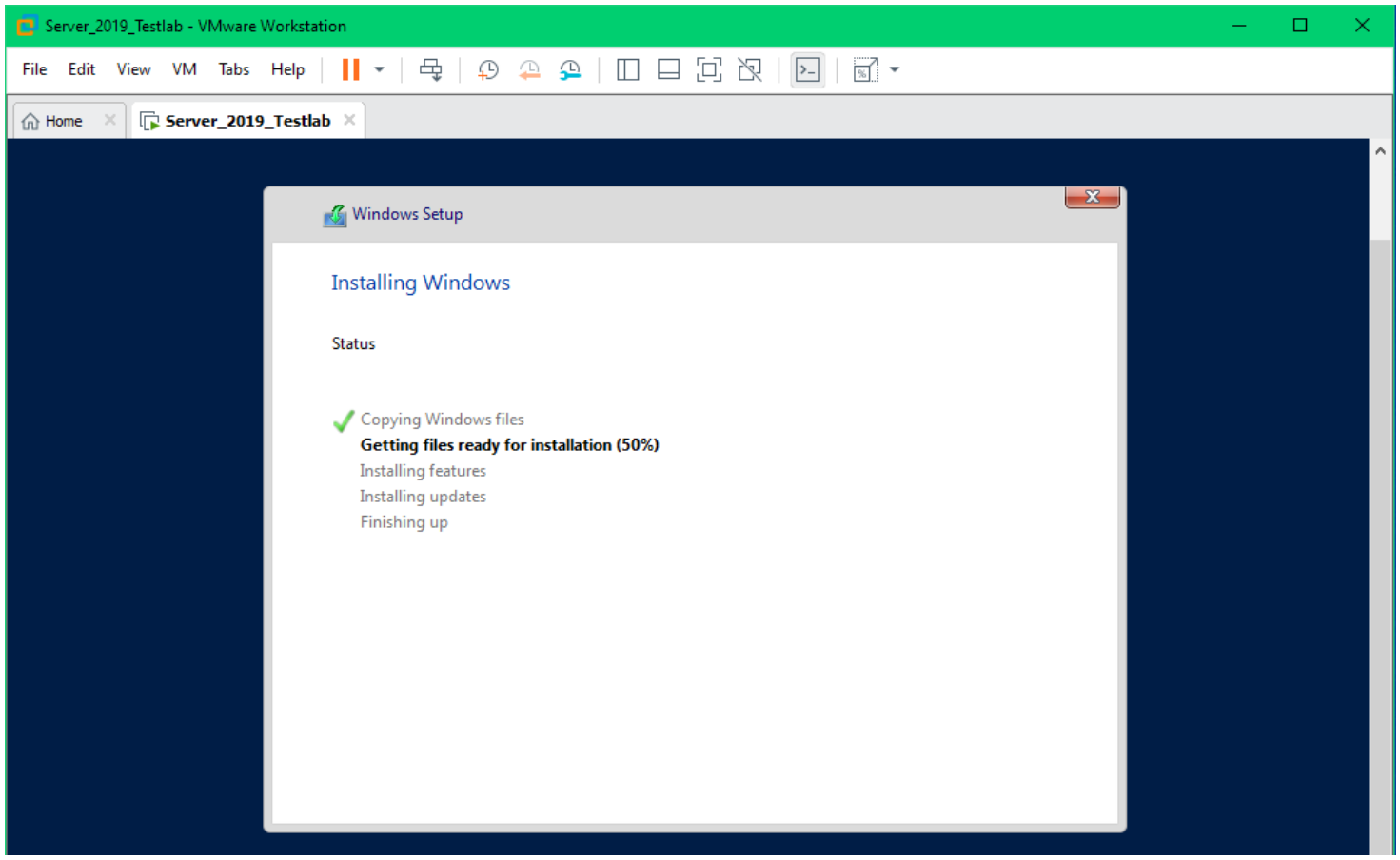
To install the OS only, you should choose the custom option. The upgrade option is upgrading or repairing an already installed Windows operating systems.

9. Select the Virtual Disk to Install the OS



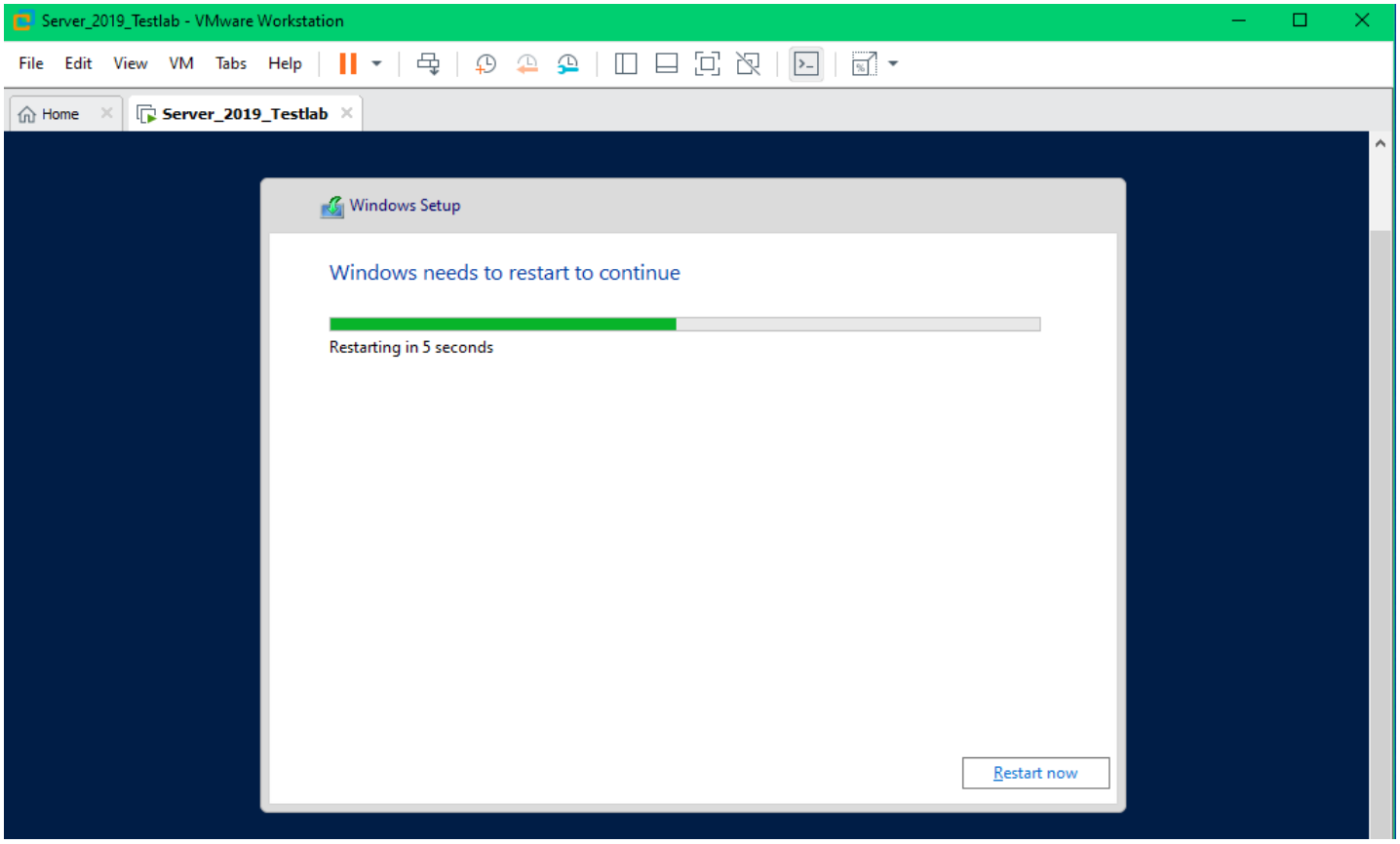
Being a brand new virtual machine we simply select the drive we created during the VM creation wizard.

10. Windows Setup – Installation of the OS



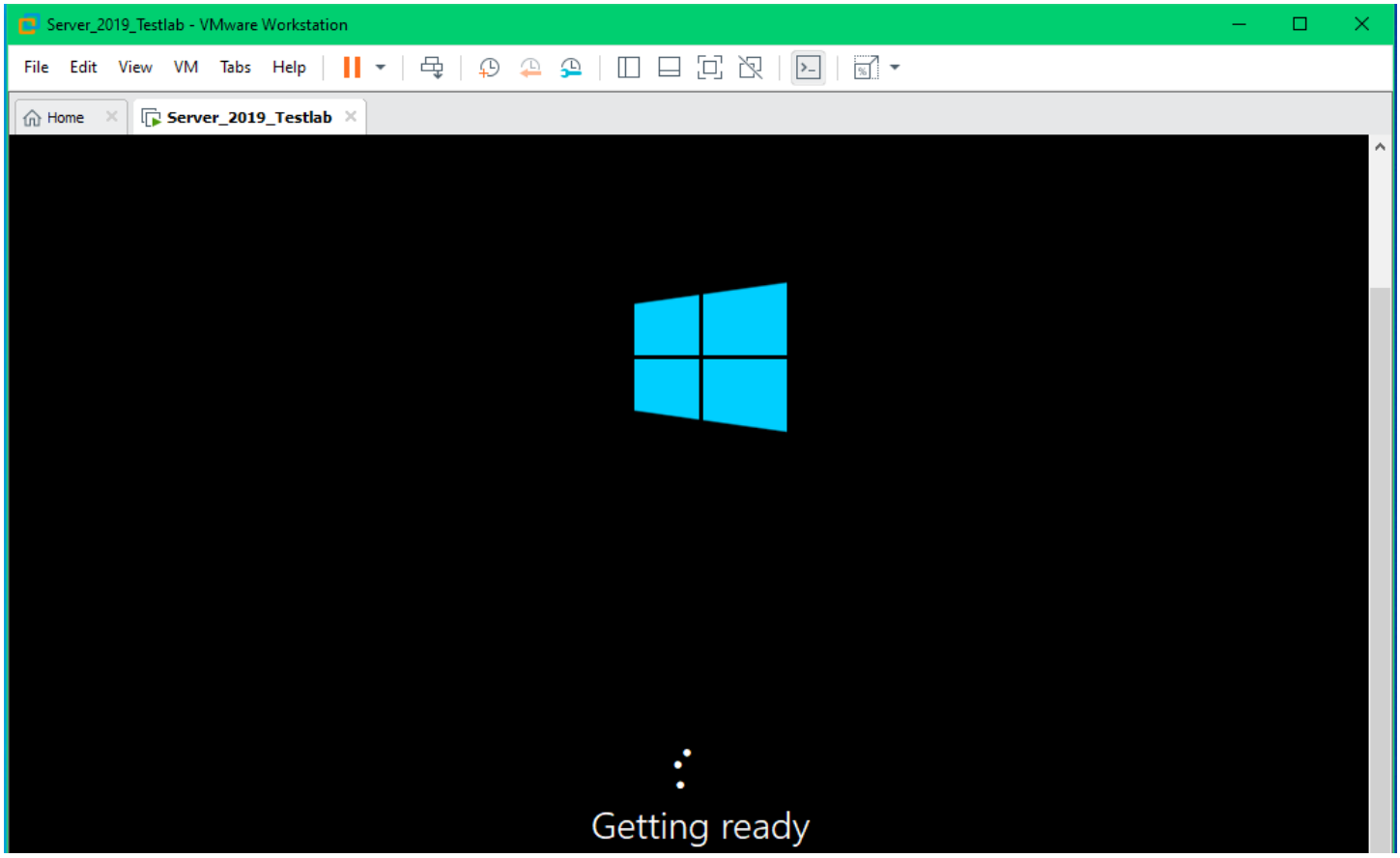
The Windows setup wizard will now install the OS of your choice, this takes approximately 5 minutes.

12. Restart to Continue the Installation



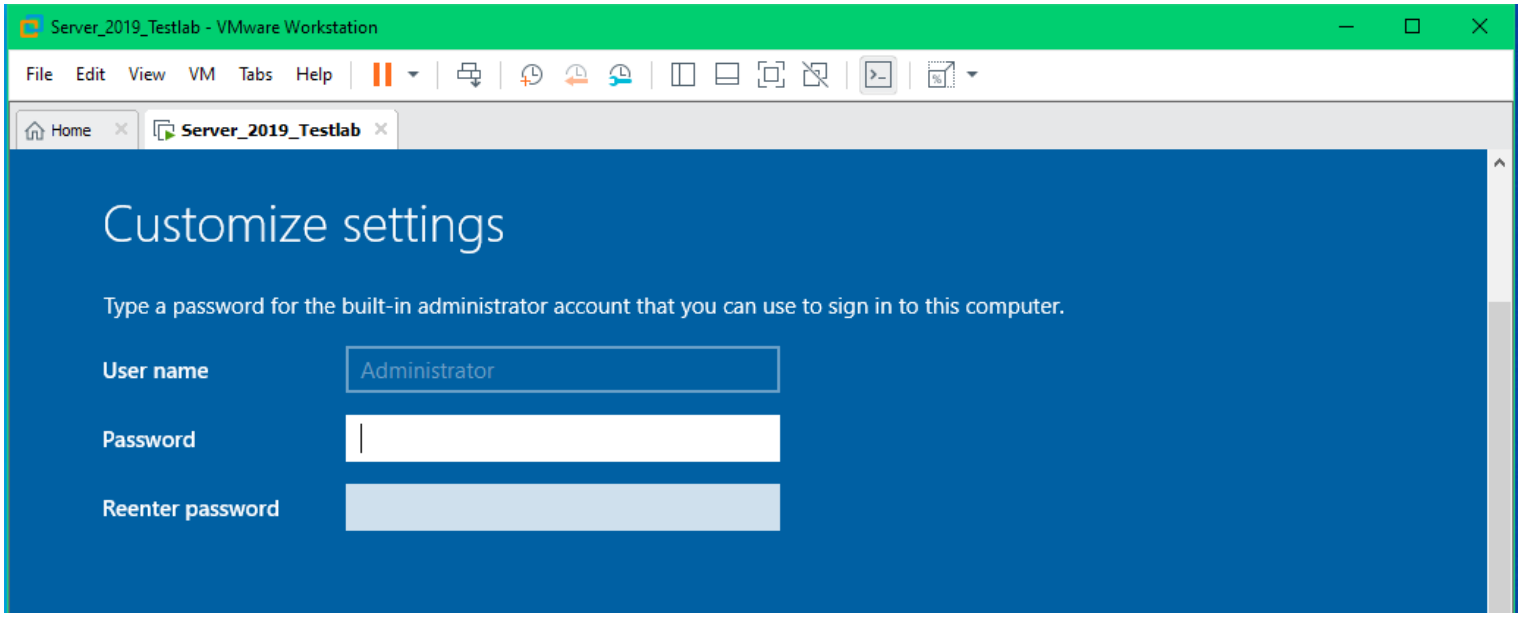
The setup wizard must restart the virtual machine to continue the install, click restart now or simply wait.

13. After Restart



The setup wizard resumes getting things ready in order to complete the installation of the guest operating system. Once complete, the OS will continue with its configuration at first-boot.

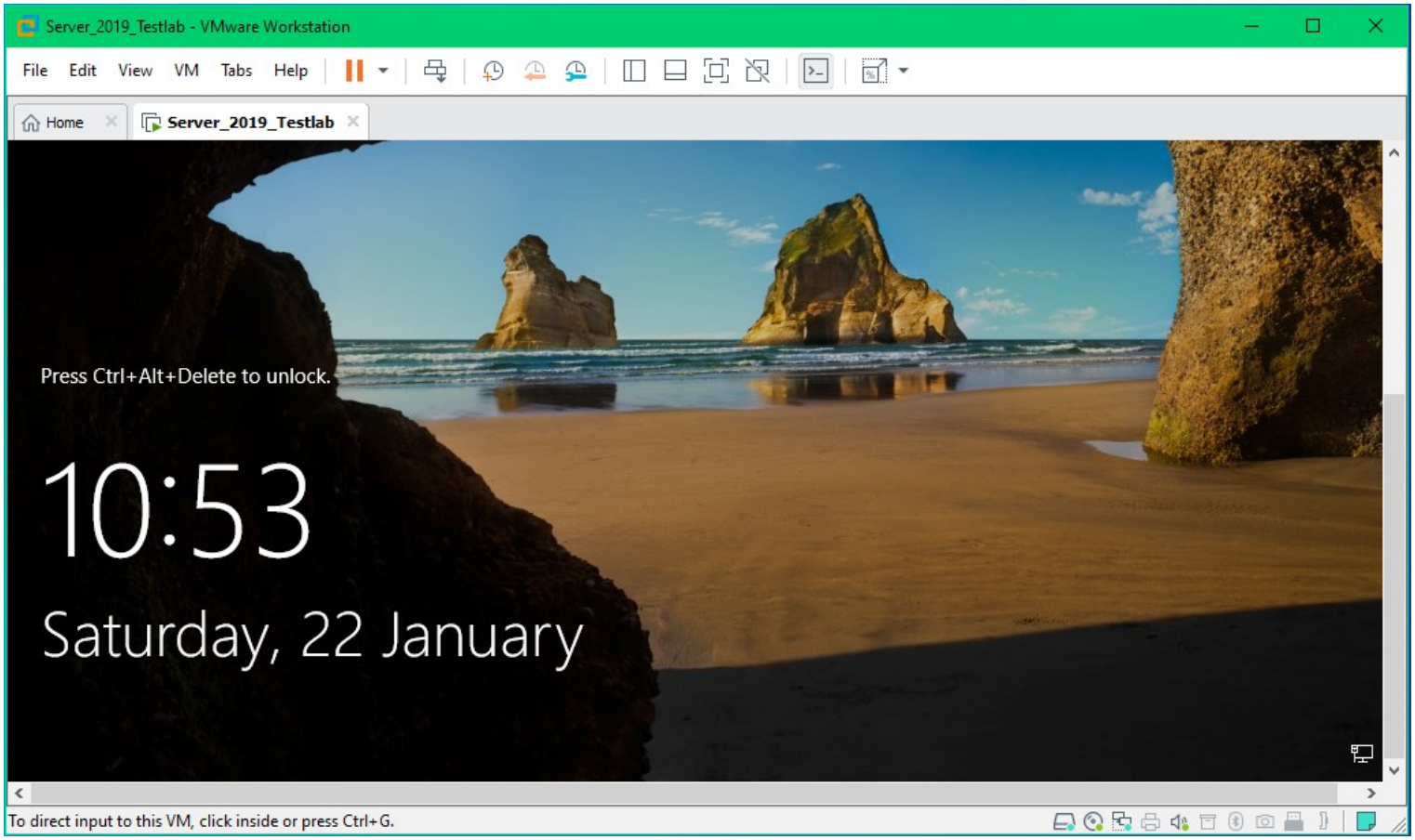
14. Set a Password for the Built-in Administrator account



You are now required to create a password for the built-in local Administrator account which we will use to login to the guest operating system.

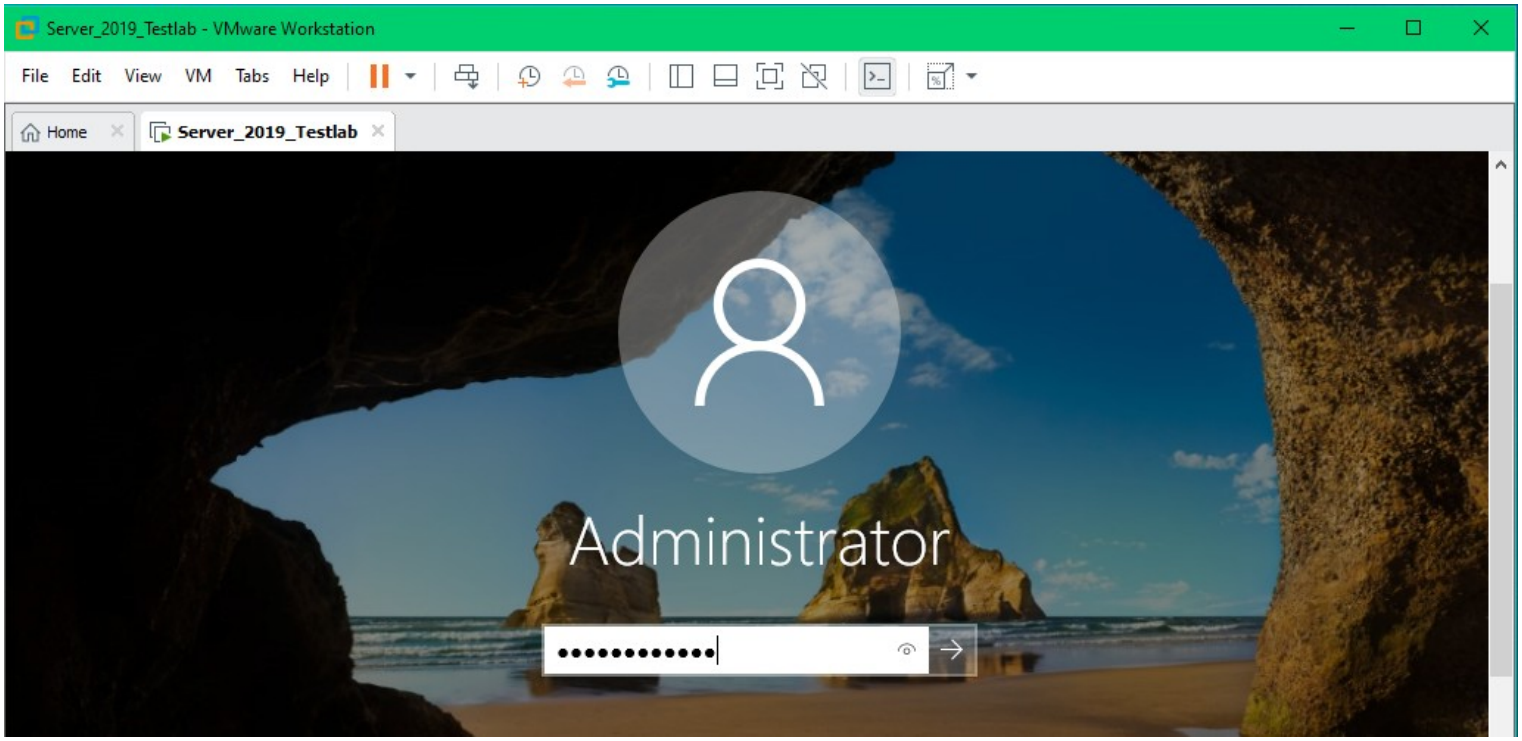
Note. Ideally for locally hosted virtual machines a simple 8 – 12 character password consisting of Upper & Lower case letters, numbers and a special character will be sufficient and make your life easier. However, if you are planning on running this in an active enterprise environment, I would suggest a much stronger password or passphrase, at least 20 characters in length, generated in an algorithmically secure fashion.

15. Server 2019 Lock Screen



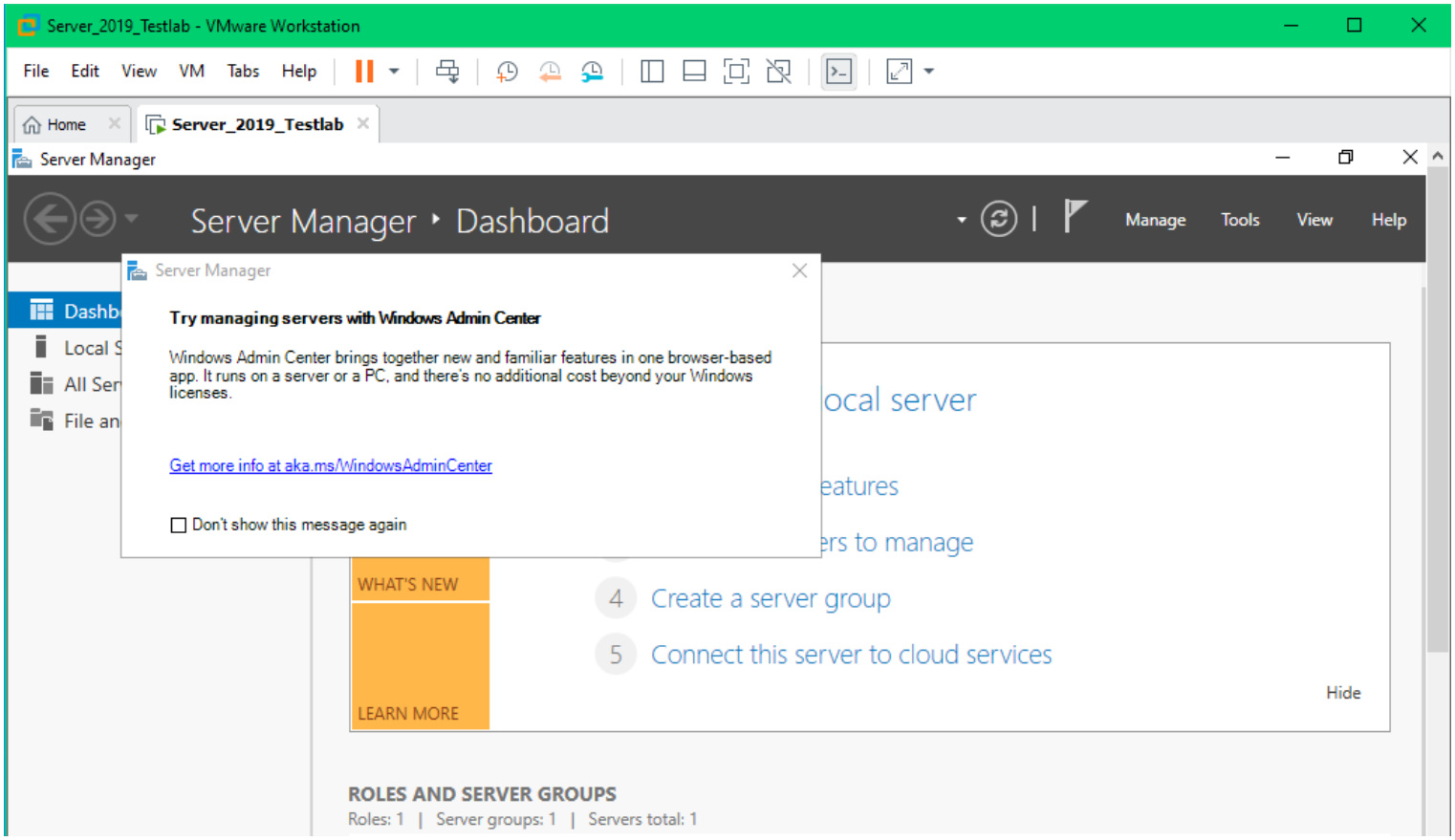
To login to the OS you will need to enter the keyboard combination; Control + Alt + Insert to send the keyboard command into the virtual machine. If this fails, you may need to use the keyboard combination; Control + Alt + Delete. Which will launch the task manager in your host as well as the guest OS.

16. Login for the First Time



Login with the password you created during the final stages of the install as the user Administrator.

17. First Login



Upon successful login the OS will boot for the first time, as I have just installed Windows Server 2019. The Server Manager application has launched which is standard by default. You may notice the display doesn't quite fit properly and/or your mouse and keyboard don't work too well.

To fix these issues, read on.

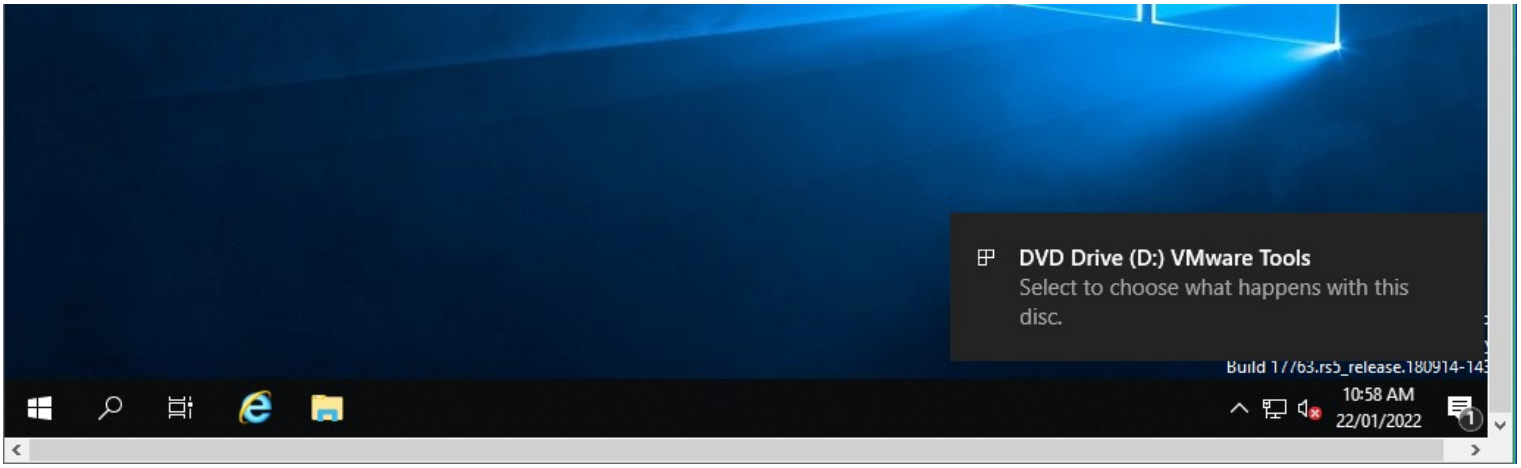
VMware Tools Installation

VMware Tools is a set of features, services and components that are designed to enhance user experience and the overall seamless usability of VMware products. Some of the included features are; enhanced drivers for virtual machines and copy & paste functionality between guest and host.

18. Install VMware Tools

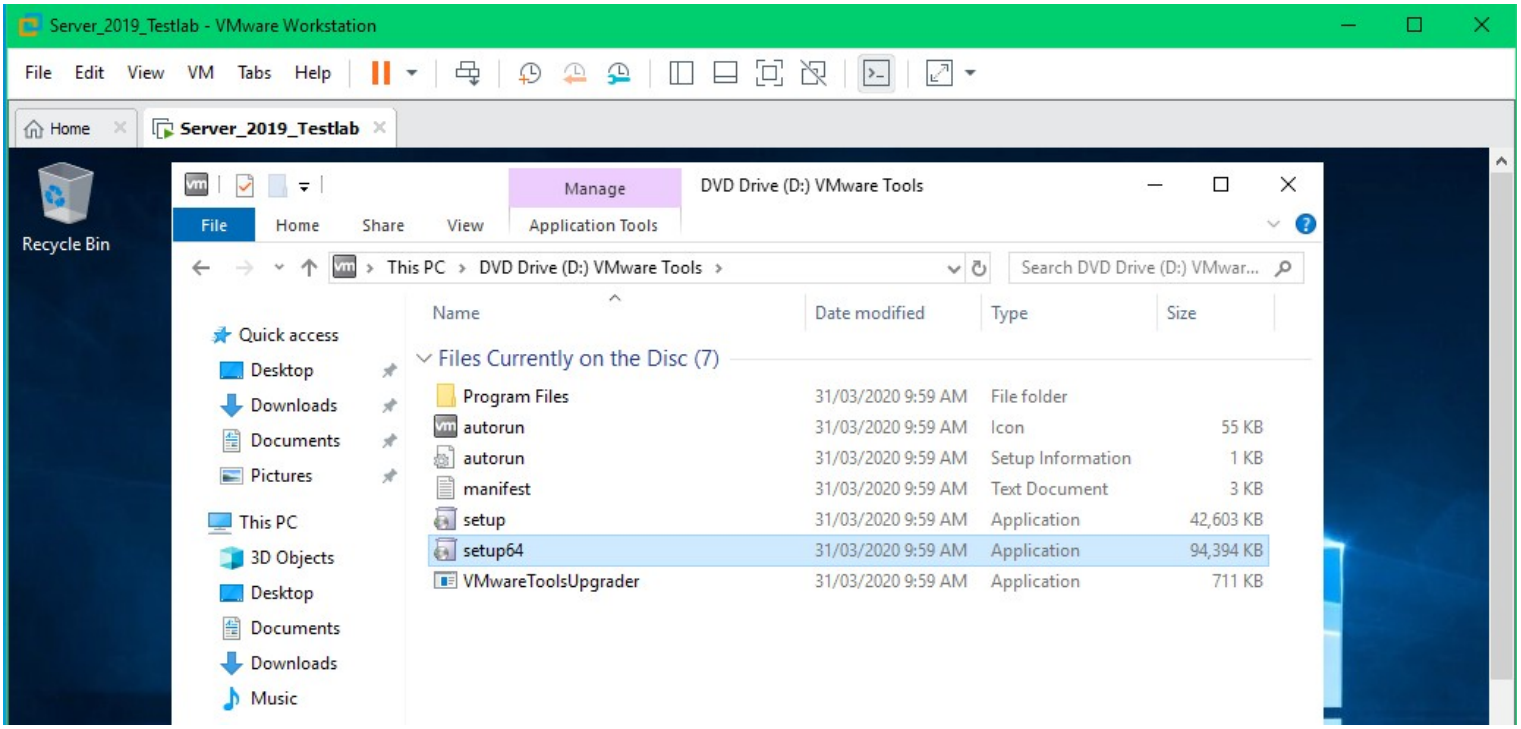
To install VMware Tools (for Windows) you will need to select the VM menu from the toolbar and simply click on Install VMware Tools. This will mount the VMware Tools ISO file to our virtual CD/DVD drive.

19. VMware Tools Disk Inserted



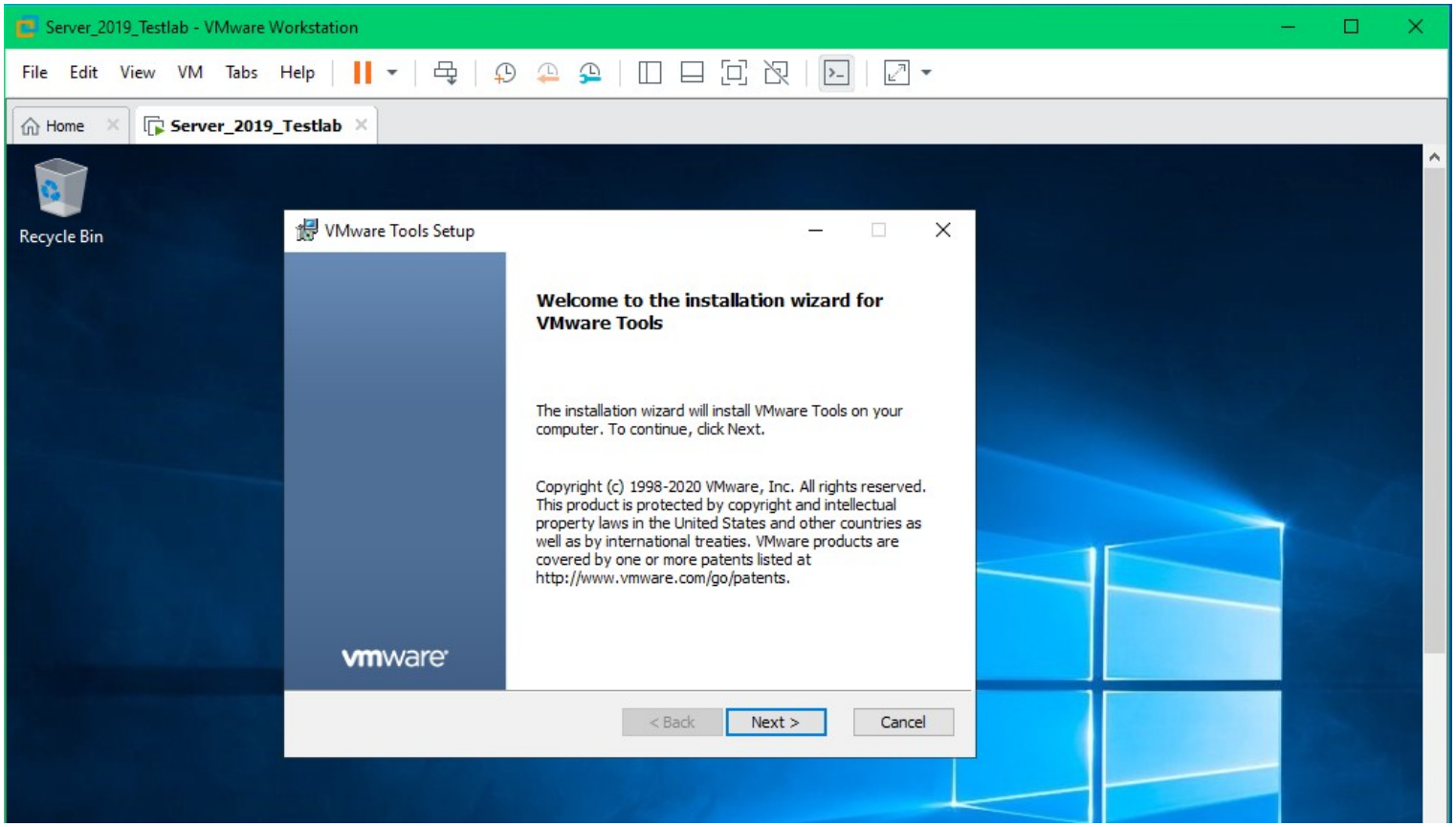
Once the VMware Tools disk image is inserted a prompt will be displayed in the bottom right corner simply click the prompt. Alternatively you can navigate to the directory in Explorer.

20. VMware Tools DVD Directory Listing



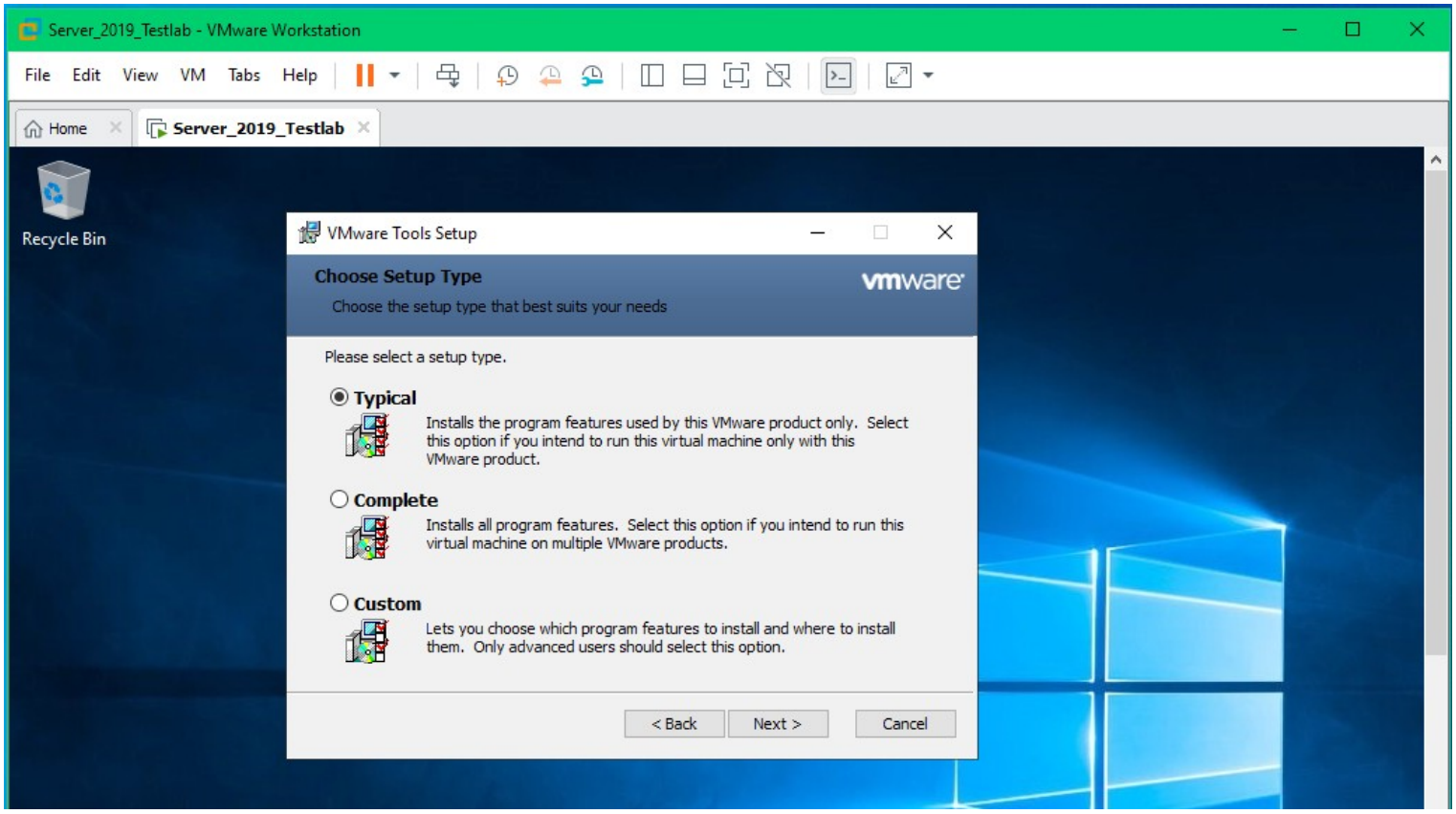
We are now presented with the directory containing the setup files, as this is a 64-bit version of Windows I will be using the setup64 executable.

21. VMware Tools Installation Wizard



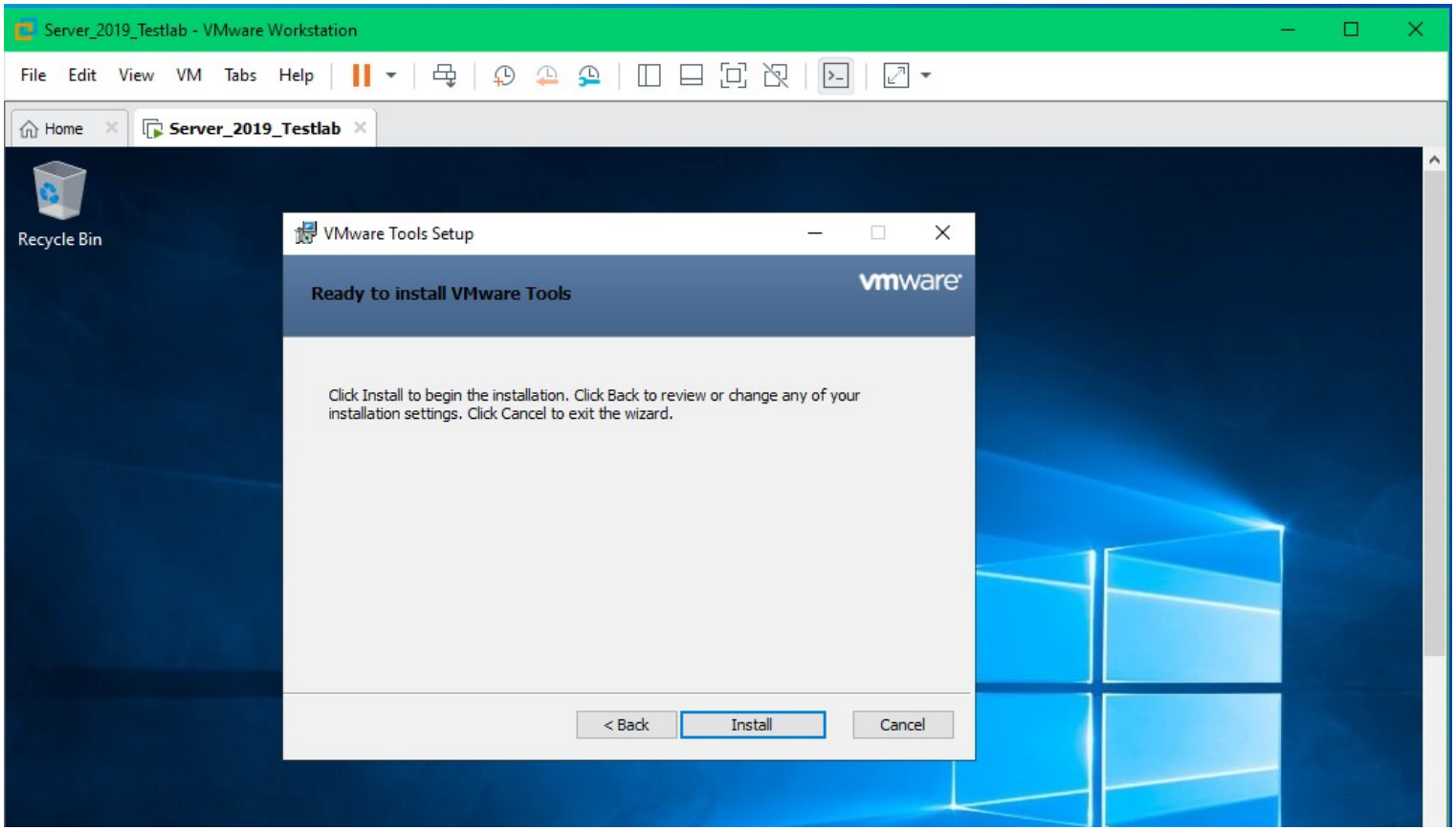
By clicking the next button the installation wizard will get the process started.

22. Select Type of VMware Tools Installation



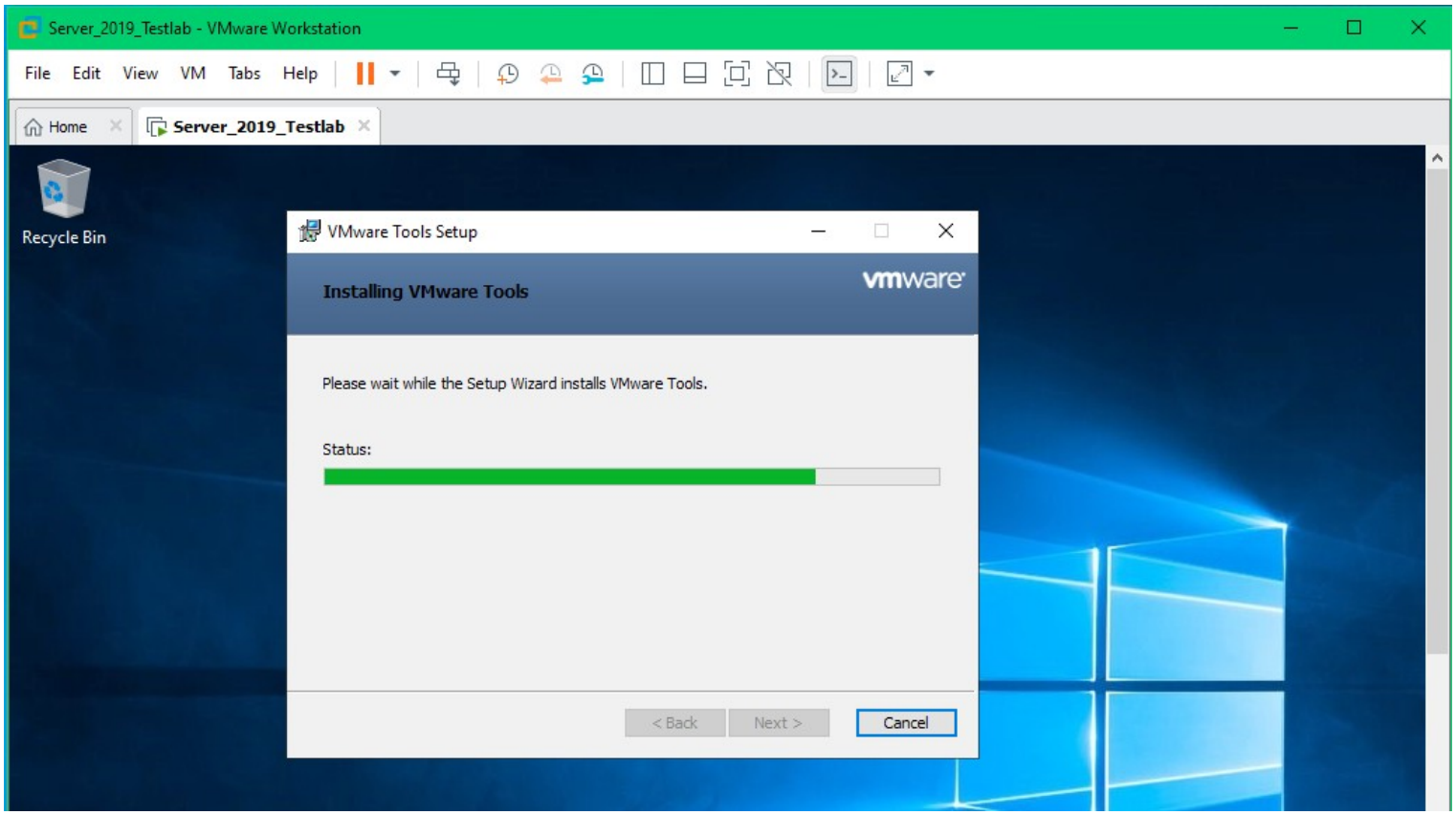
For a general-use virtual machine, I find a typical installation to satisfy my usage requirements.

23. Ready to Install VMware Tools



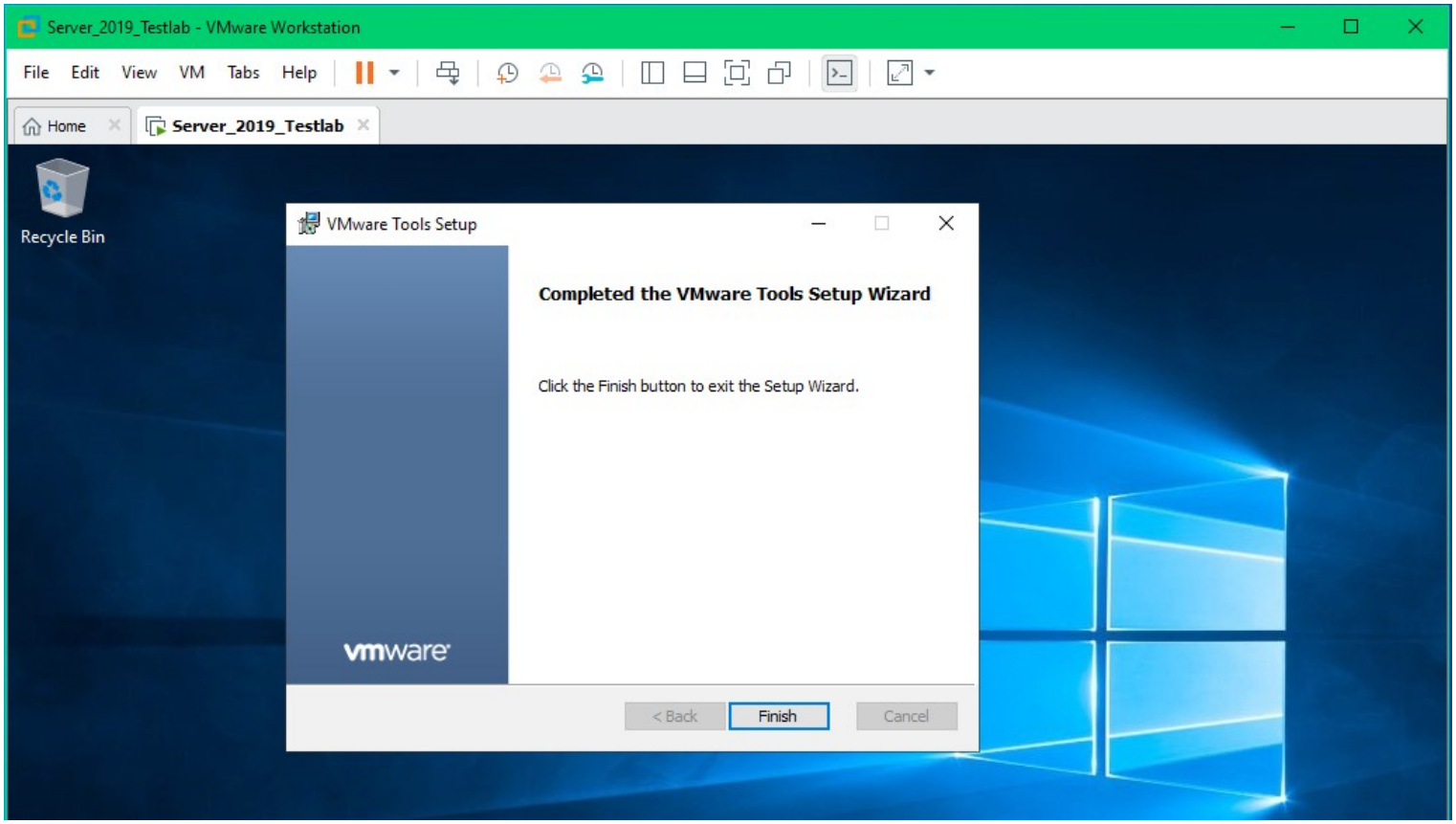
By clicking Install, the installation process will commence and install VMware Tools to the guest OS.

24. Installing VMware Tools



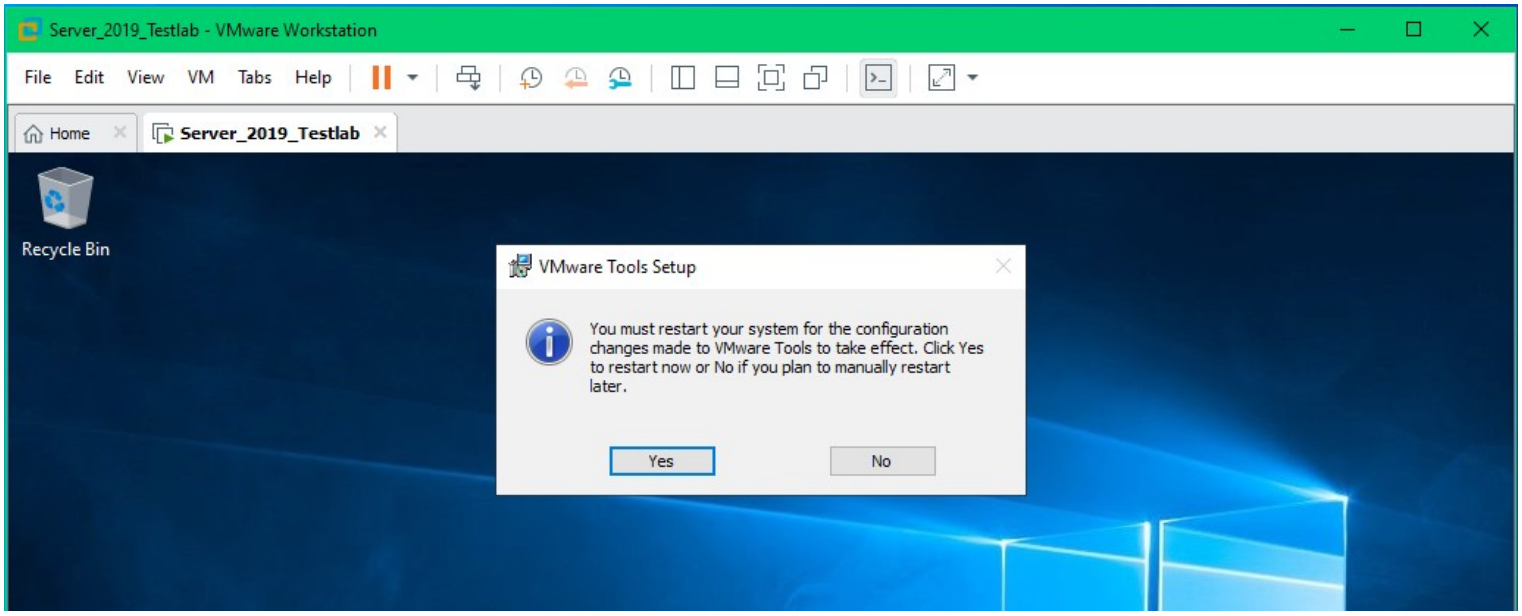
The installation process doesn’t take long to complete.

25. Completed VMware Tools Setup Wizard



The installer has completed the installation of VMware Tools, simply click finish to exit. Do NOT restart!

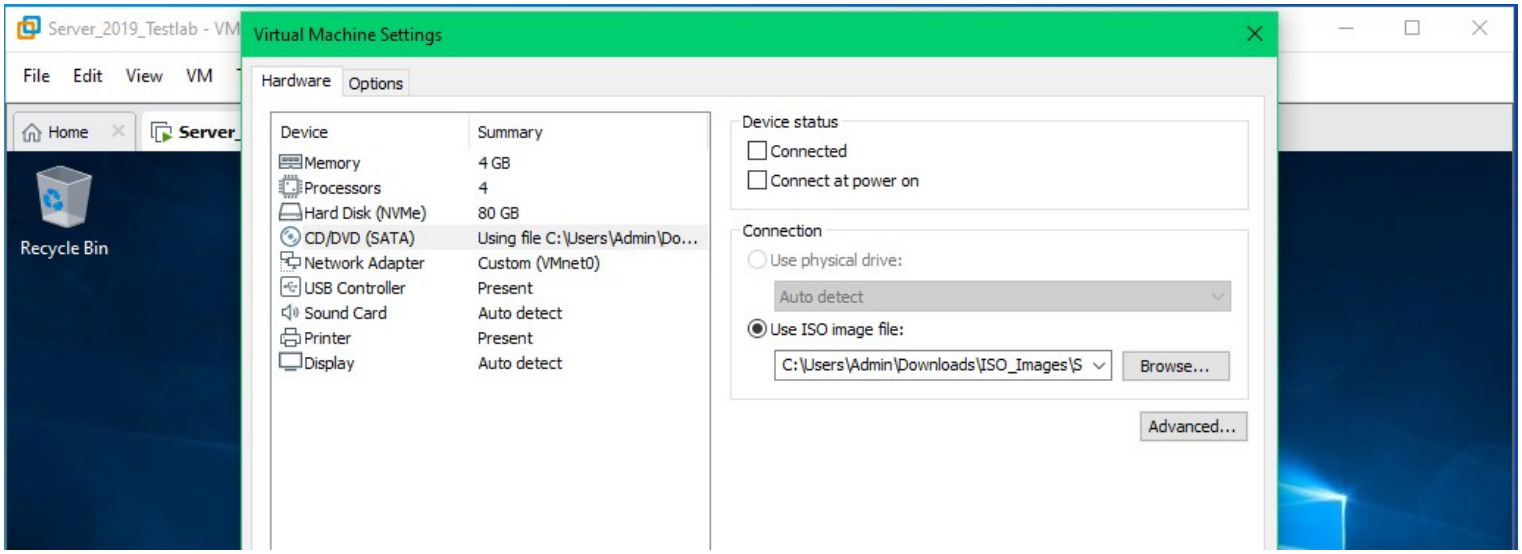
26. Do **NOT** Restart your Guest OS



Prior to restarting the guest OS we have to disable our virtual CD/DVD drive being connected at boot. This is to prevent the VM from detecting 2 different boot-able drives at the time of boot, leading to issues.

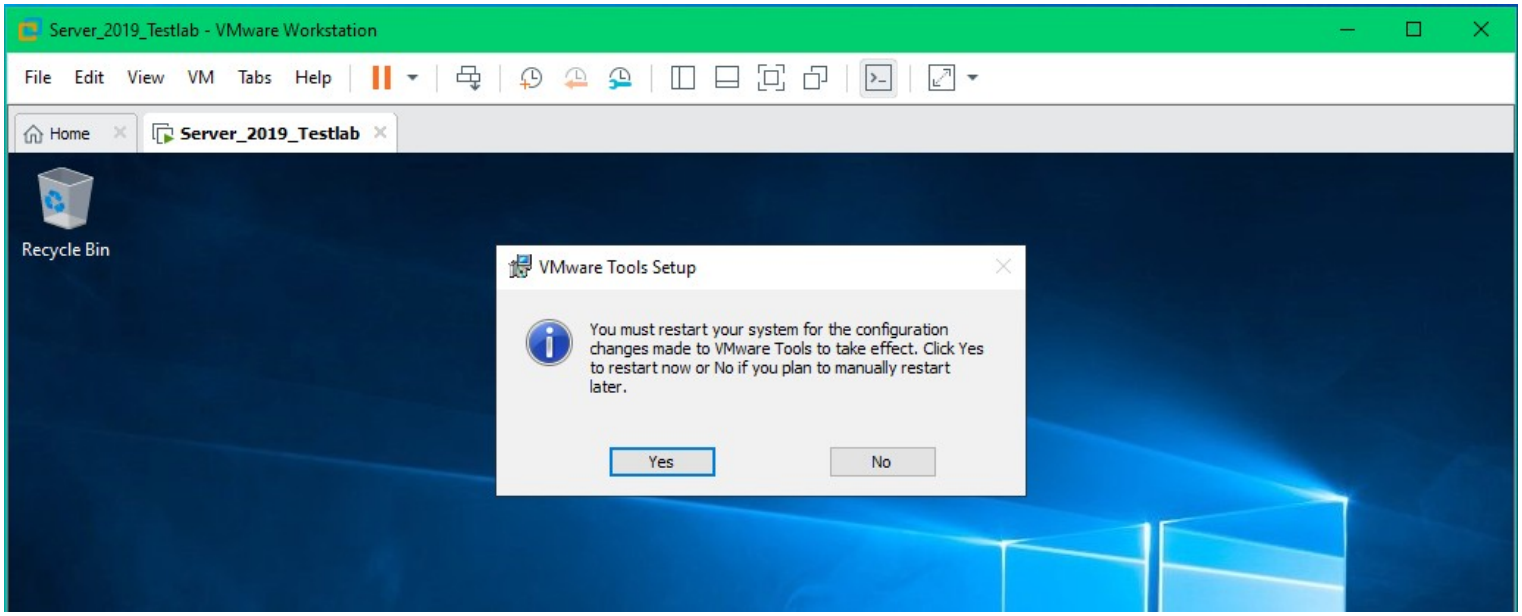
To do this you must simply head to the VM menu at the top of the window and select the bottom option Settings.

28. Virtual Machine Settings – CD/DVD



By deselecting the connected at power-on check box, the virtual drive with the ISO image file will not be mounted to our virtual machine.

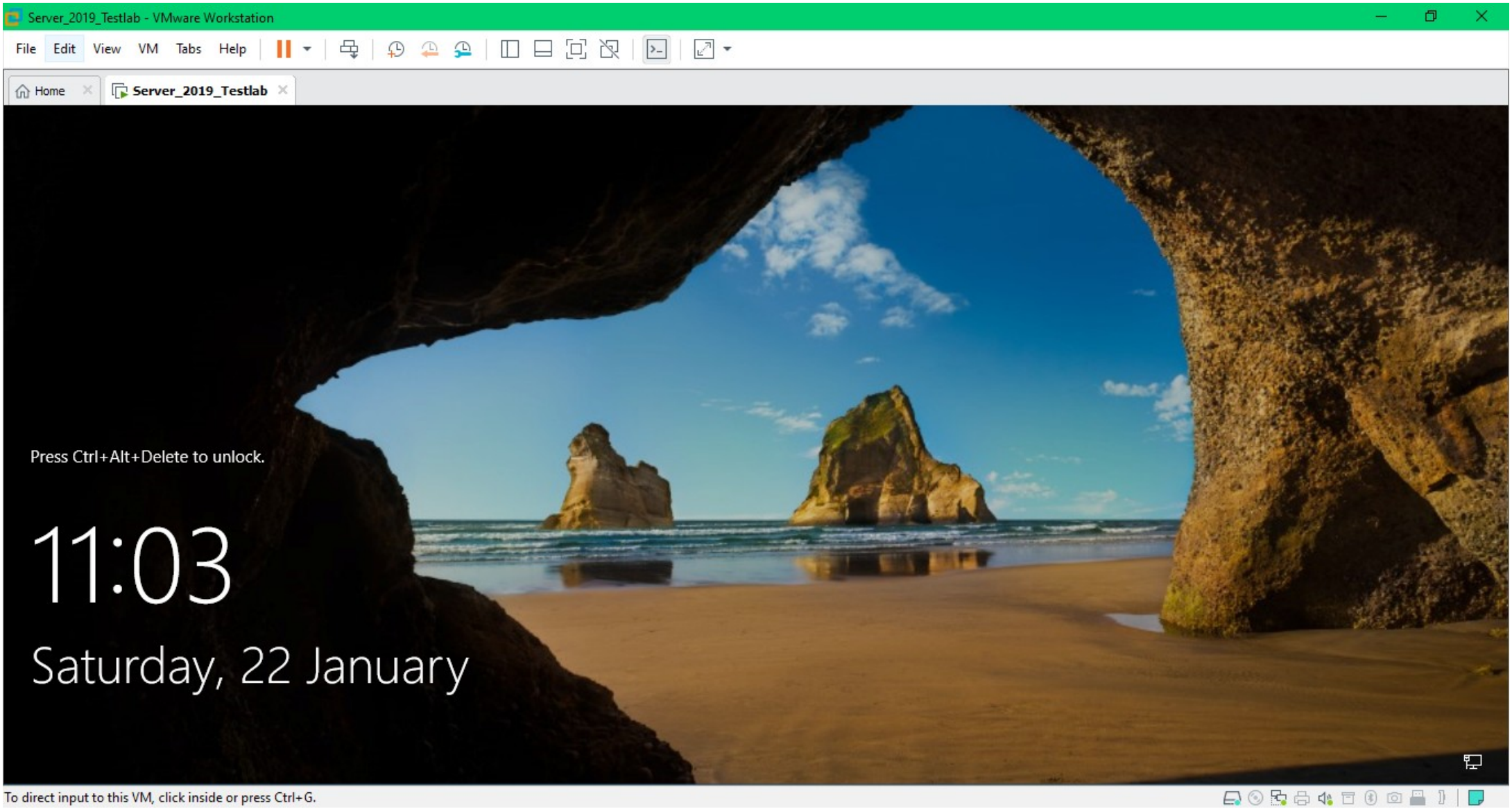
29. Restart the Guest OS



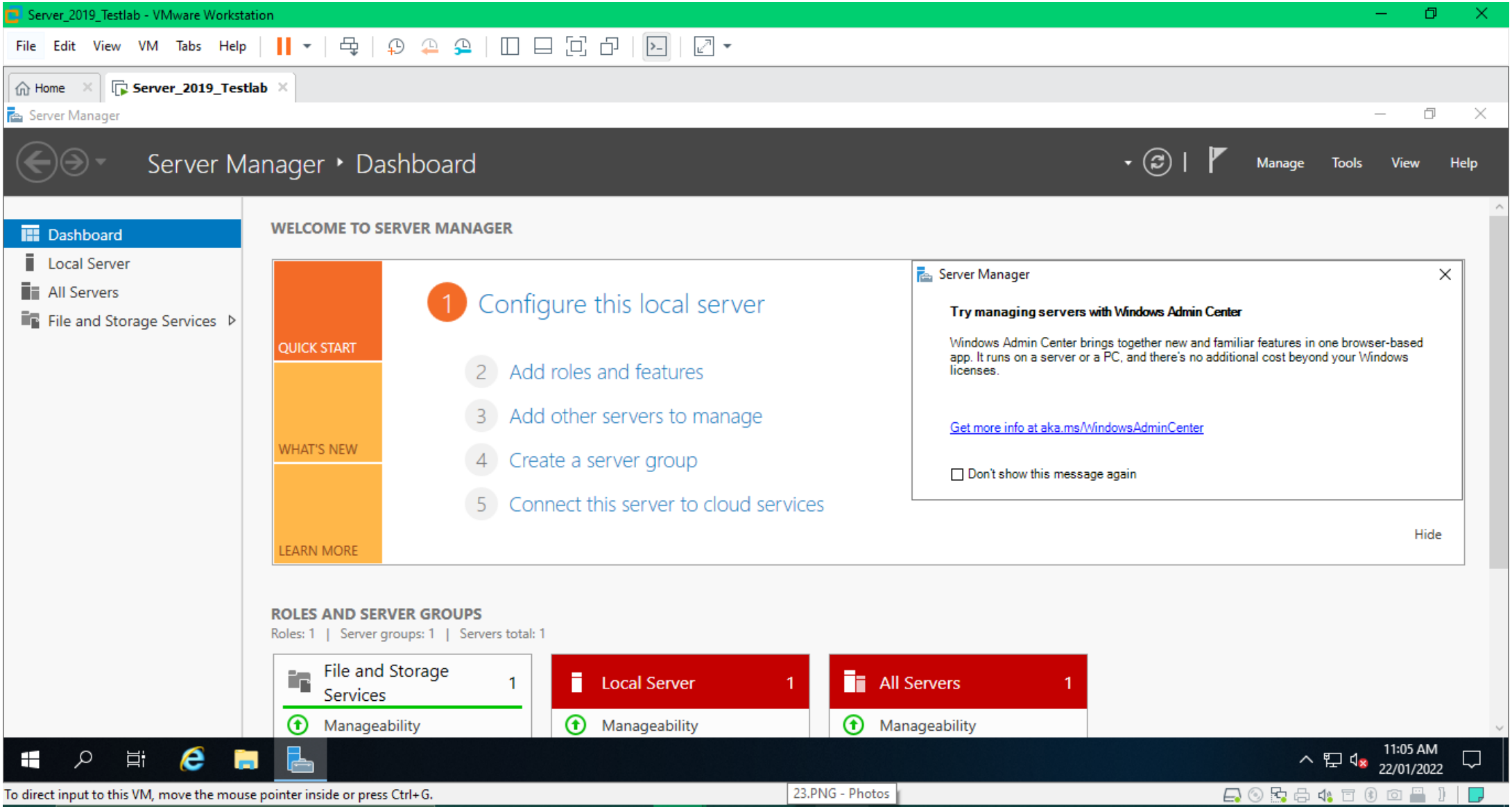
Click Yes to restart the guest OS to allow for the changes made by the VMware Tools installer to take effect.

After restarting the guest OS, you will be presented with a much more user friendly operating system then before VMware Tools was installed. The first thing you will probably notice is that the guest OS's screen resolution will automatically adjust and VMware will fit the whole desktop window nicely in your display.

Windows Server 2019 Complete with VMware Tools



You can now see that the desktop of the virtual machine fits the display of my laptop perfectly with all 4 corners of the desktop viewable without the need to scroll up,down, left or right.



This time when the Guest OS successfully boots and launches Windows Server Manager, it fits nicely in the display with the guest OS taskbar visible within the virtual machines display.

Snapshot Your New VM

A virtual machine snapshot is an image of the data and state of the VM at a given point in time. This creates a type of restoration point where the machine can safely be reverted to a prior state. The data contained within the image will include; all files from disk and in memory as well as any other supported storage devices attached to the VM.

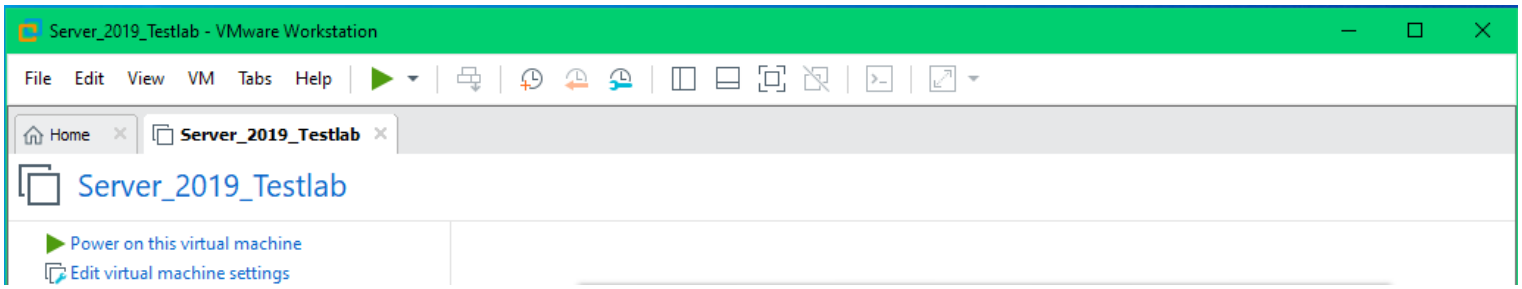
Taking snapshots is highly beneficial as it allows for a quick and easy revert should the need arise. There are several reasons which you may choose to revert to a prior snapshot including;

- Practice – Reverting a VM is a quick and easy way to practice setting up services, roles and features on Windows Server OS's.
- Errors - Sometimes things go wrong and virtual machines stop working correctly. Having a snapshot to revert back to, is an easy way to fix the error and get the VM functioning correctly.

1. Power-off Virtual Machine

To take a snapshot of your virtual machine it is best that the guest OS is shutdown rendering the VM in the powered-off state. I believe you can take a snapshot with the virtual machine in any state however, for our purposes it is best to power off the machine first.

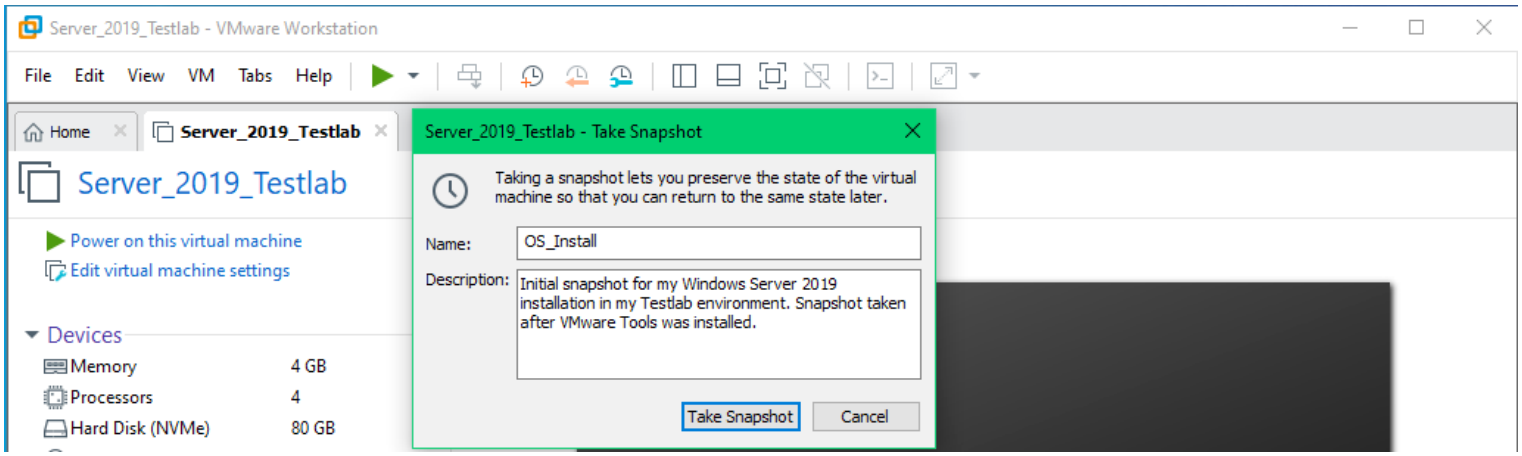
2. VMware Home Screen



In the top toolbar within VMware select the Take Snapshot button.

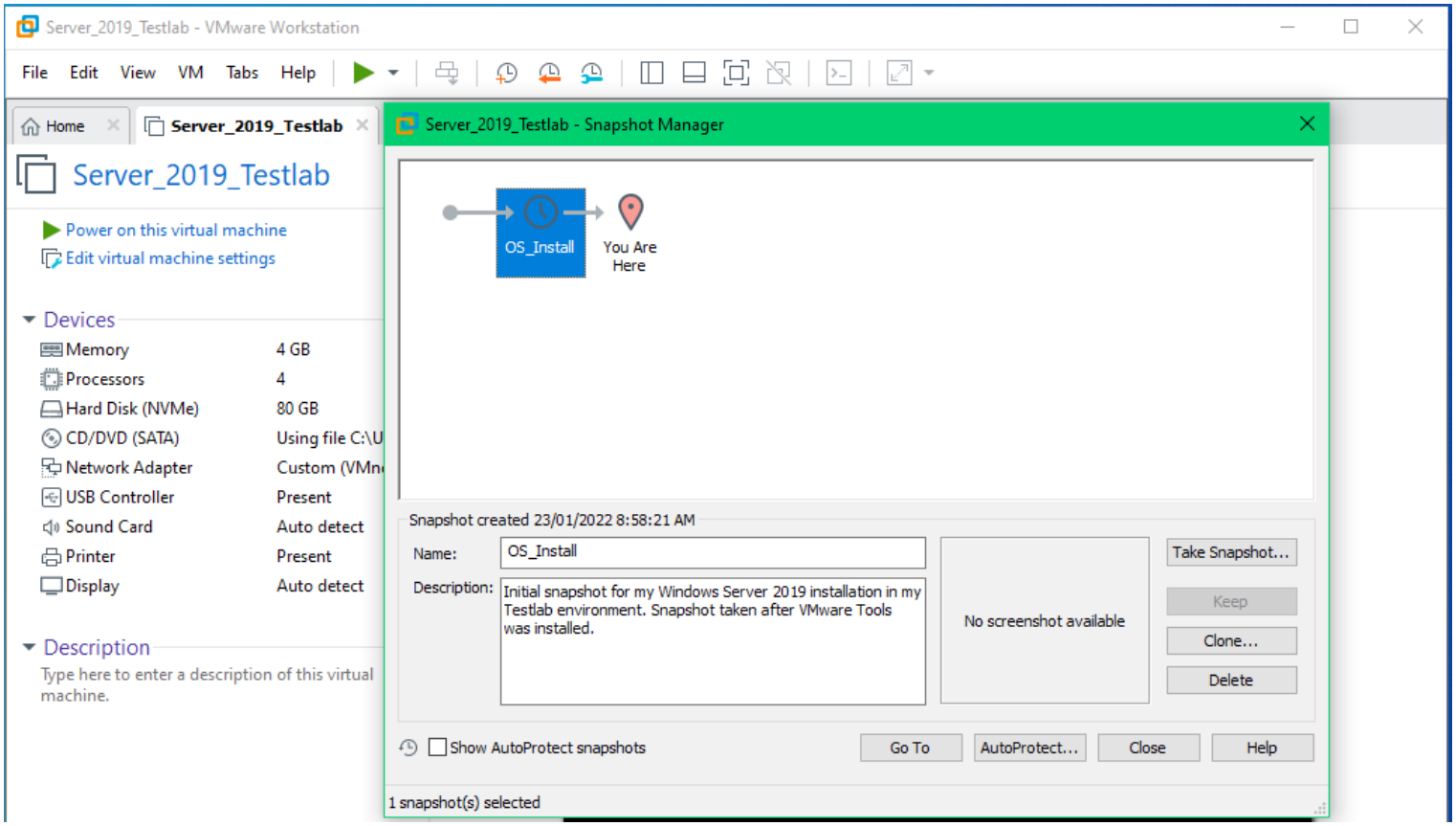
Select the clock symbol with the + icon (left) ----->

3. Name The Snapshot



Add a meaningful name and a description if you desire. I simply named this snapshot as OS_Install and gave it a brief description.

4. Review The Snapshot



By selecting the snapshot manager icon from the toolbar, you can view all snapshots for the virtual machine.

Snapshot Manager is the clock icon with the spanner pointing left (Right) <------

To revert to a snapshot you can simply click on the clock icon with the arrow pointing to the left, this will revert the virtual machine to the last snapshot.

Summary

In this tutorial we have covered how to create a new virtual machine in VMware Workstation and installed Windows Server 2019 a guest operating system. We then briefly covered the function and purpose of VMware Tools and the associated installation process. This is the basic process which I follow to setup standalone virtual machines, it is quite straightforward and a lot of the options you can simply accept the default options.

Note

- To return to your host – By pressing control + alt your mouse and keyboard will be returned to your host operating system.
- VMware Workstation – In this tutorial I only covered the bare basics of what you can accomplish in this product. I will cover a lot more in tutorials to come.
- Networking – I have configured the network adaptor to be bridged with my hosts physical wireless adaptor, this means the machine will be allocated an IP address from my home networks DHCP server.
- Same Process - This tutorial will work for almost all guest OS types (Linux and Windows). Although the options and configuration settings may be slightly different, the overall process remains the same.

Disclaimer

I am in no way affiliated with VMware, I simply like Workstation and happen to have a valid product key.

Check my tutorials page for more!
https://m4773l.github.io/tutorials